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CAPSTONE PROJECT REPORT

Report 3 – Software Requirement Specification

OptiPack AI – AI-Assisted Omnichannel Order Fulfillment and Packaging Optimization (FA26SE036)

– Ho Chi Minh City, October 2026 –

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I. Record of Changes

Date	A* M, D	In charge	Change Description
24/10/2026	A	ThuanNPM	Initial SRS for Review 2: product overview, actors, use cases, system functional overview, functional requirements of 12 features, non-functional requirements, business rules and message list.
25/10/2026	M	ThuanNPM	Aligned with the code: two-factor authentication is optional for every role (not mandatory for Admin); order sync runs per shop interval (default 5 minutes), not every minute; manual sync is Admin only; Tiki / TikTok adapters are not registered; Dashboard Overview partly uses sample data; AI text service is Groq; packing-plan performance measured by benchmark script; mobile app noted as planned for Iteration 3.
08/11/2026	M	ThuanNPM	Use case section aligned with the redrawn use case diagrams (Figures 2–7): actor list (System Administrator, Authenticated user, Marketplace Platform, Carrier) and use case descriptions UC-01 to UC-108 now list exactly the use cases of the diagrams. Entity descriptions aligned with the logical ERD (Figure 8). Dashboard reads live data; express SLA check runs every minute; Tiki adapter status; shipping cost estimation details (3.7.1); metrics for usability and security requirements; benchmark path; regression test count updated.

*A - Added M - Modified D - Deleted

II. Software Requirement Specification

1. Product Overview

Small and medium online sellers in Vietnam sell the same products on several channels at once. Orders arrive separately from each marketplace, staff pick items from memory, pack them in whatever box is at hand and book carriers by hand. The result is mis-picked items, oversized boxes, wasted packaging material, unnecessary shipping cost and no single place to follow an order from purchase to delivery or return.

OptiPack AI is a web-based fulfillment system for one seller (one Store Owner and its warehouse team). It pulls orders from connected marketplaces into one queue, groups orders that go to the same recipient, reserves stock, guides warehouse staff to pick by bin and by oldest lot (FIFO), computes a 3D packing plan for every order (box choice, item placement, filling materials) with a step-by-step packing guide, books one carrier shipment per parcel with GHN, follows delivery, failed delivery and return to the warehouse, and handles customer return and exchange requests. Every important status change is written back to the marketplace so that end customers can track their orders.

The release covered by this SRS integrates two sales channels:

- **Lazada** – real marketplace connected through the Lazada Open Platform (OAuth; a scheduler runs every minute and synchronizes each shop when its configured interval has passed, default 5 minutes). OptiPack AI does not write order status back to Lazada (read-only by design).
- **AURELLE** – a fashion web store built by the team (customer website + Open API that follows the Lazada API style, HMAC-signed requests and webhooks). AURELLE plays the role of a second marketplace and lets the team demonstrate the full loop: order → fulfillment → customer tracking → return / exchange.
- **TikTok Shop, Shopee and Facebook Marketplace – out of scope of this release because of partner API access constraints (see Form BM03); the TikTok Shop adapter file remains in the repository but is not registered. Tiki – an adapter based on a Tiki seller Inhouse app (dropship orders) was added on 31/10/2026, after the Review 2 scope was fixed; it is registered in the back end but is not part of the Review 2 demo and is not specified further in this SRS.**

The context diagram below shows the boundary of OptiPack AI and the external entities it exchanges data with.

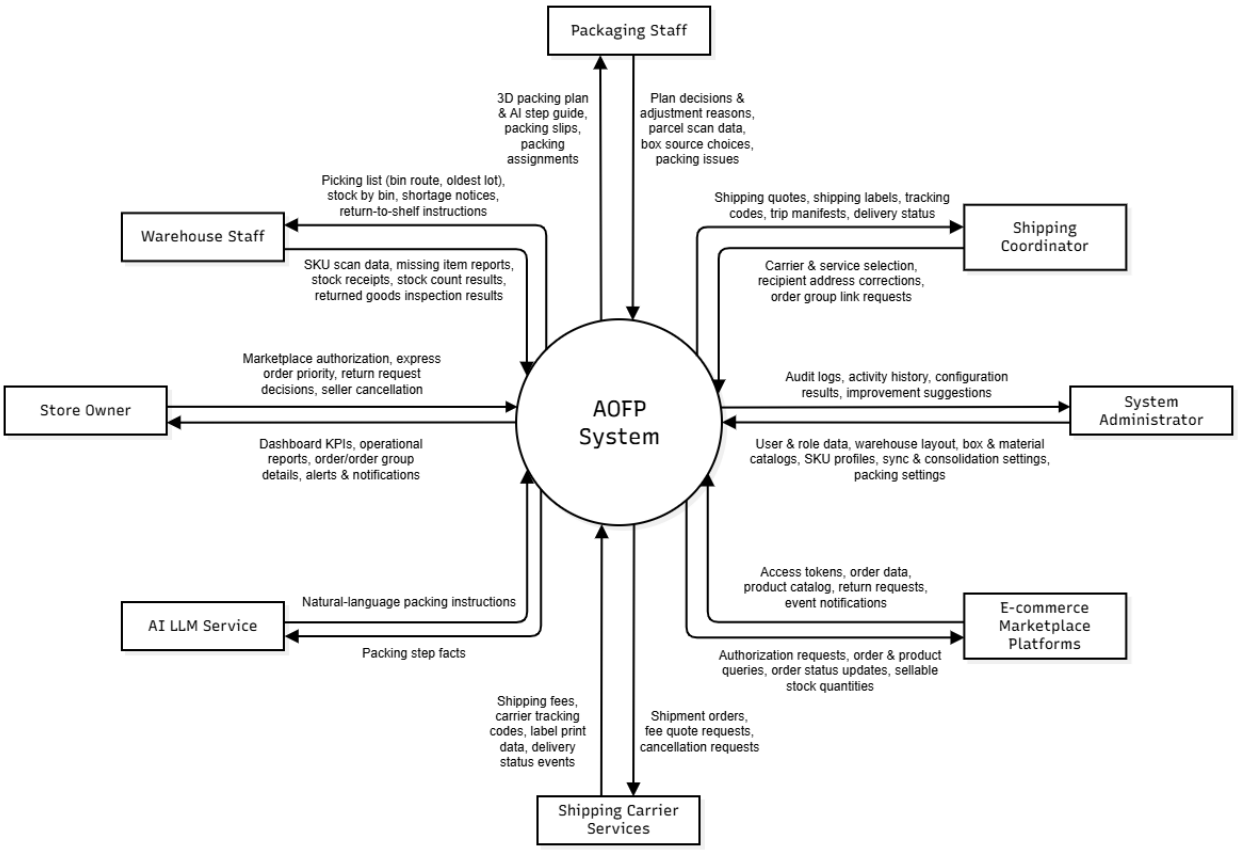


Figure 1. Context diagram of OptiPack AI

External entity	Type	Data exchanged with OptiPack AI
Lazada Open Platform	External system	OAuth tokens, orders and order items, product catalog (seller SKUs)
AURELLE Website / Open API	External system (team-built marketplace)	Orders, order-changed / product-updated / return-requested webhooks, return requests, order status and return status written back, available stock pushed every minute
GHN (Giao Hàng Nhanh)	External system (carrier)	Shipping fee quotes, shipping orders per parcel, tracking codes, status webhooks, A5 labels
Google Identity	External system	OAuth 2.0 login for staff whose e-mail already exists in the system
Gmail SMTP	External system	Account, temporary password, password reset and lock notification e-mails
AI text service (Groq API, OpenAI-compatible)	External system	Packing-guide wording for a parcel (no personal data)
Staff (5 roles)	People	Operate the web portal: configuration, picking, packing, shipping, returns
Customer (AURELLE)	People	Places orders, tracks delivery, requests return / exchange on the AURELLE website

2. User Requirements

2.1 Actors

#	Actor	Description
1	System Administrator (Admin)	Full access. Manages user accounts, roles and leave, connects / disconnects marketplace shops, configures order sync, warehouses, internal SKUs, the

#	Actor	Description
		packaging catalog and carriers. In the use case diagrams the System Administrator also inherits the use cases of the Store Owner, Warehouse Staff, Packaging Staff and Shipping Coordinator.
2	Authenticated user	Generalization of every staff role (Admin, Store Owner, Warehouse Staff, Packaging Staff, Shipping Coordinator): logs in, changes or resets the password, sets up optional two-factor authentication, manages the own profile and views notifications.
3	Store Owner	Business owner. Monitors orders and order groups, sets express priority, cancels AURELLE orders as seller, splits / merges groups, reviews return requests and creates replacement orders, configures packing and shipping settings, sets staff leave, views packing reports.
4	Warehouse Staff	Picks items for assigned order groups, reports missing items, receives stock by lot, adjusts and transfers stock, confirms SKU packaging profiles, prints labels, receives returned parcels and inspects returned items. The back end also allows Warehouse Staff to pack parcels.
5	Packaging Staff	Reviews, approves, adjusts or rejects packing plans, packs parcels (scan, seal, weigh), reviews held parcels, decides partial shipment when items are missing.
6	Shipping Coordinator	Compares carrier quotes, creates shipments (single group or same-recipient batch trip), schedules pickup, corrects delivery addresses, records express self-delivery, links / unlinks, splits or merges order groups.
7	Customer	End buyer on the AURELLE website. Registers, browses products, places and cancels orders, tracks delivery and requests return or exchange.
8	AURELLE Web Admin	Administrator of the AURELLE website. Manages storefront categories, products, variants and images, and Open Platform developer apps (app key / secret).
9	Marketplace Platform	External sales channels connected through their Open API: Lazada Open Platform (orders and catalog of the connected shop) and AURELLE Open Platform (orders, catalog, return requests and webhooks; receives status and stock written back by OptiPack AI).
10	Carrier	External carrier system (GHN) that quotes fees, creates shipping orders and reports delivery status.
11	System Scheduler	Time-based jobs inside OptiPack AI: order and catalog sync, picker assignment, packing-plan computation, shortage / blocked recheck, status write-back, stock push, return pull, SLA alerts.

2.2 Use Cases

2.2.1 Diagram(s)

Use cases are grouped into packages so that each diagram stays readable. One diagram is drawn per package below.

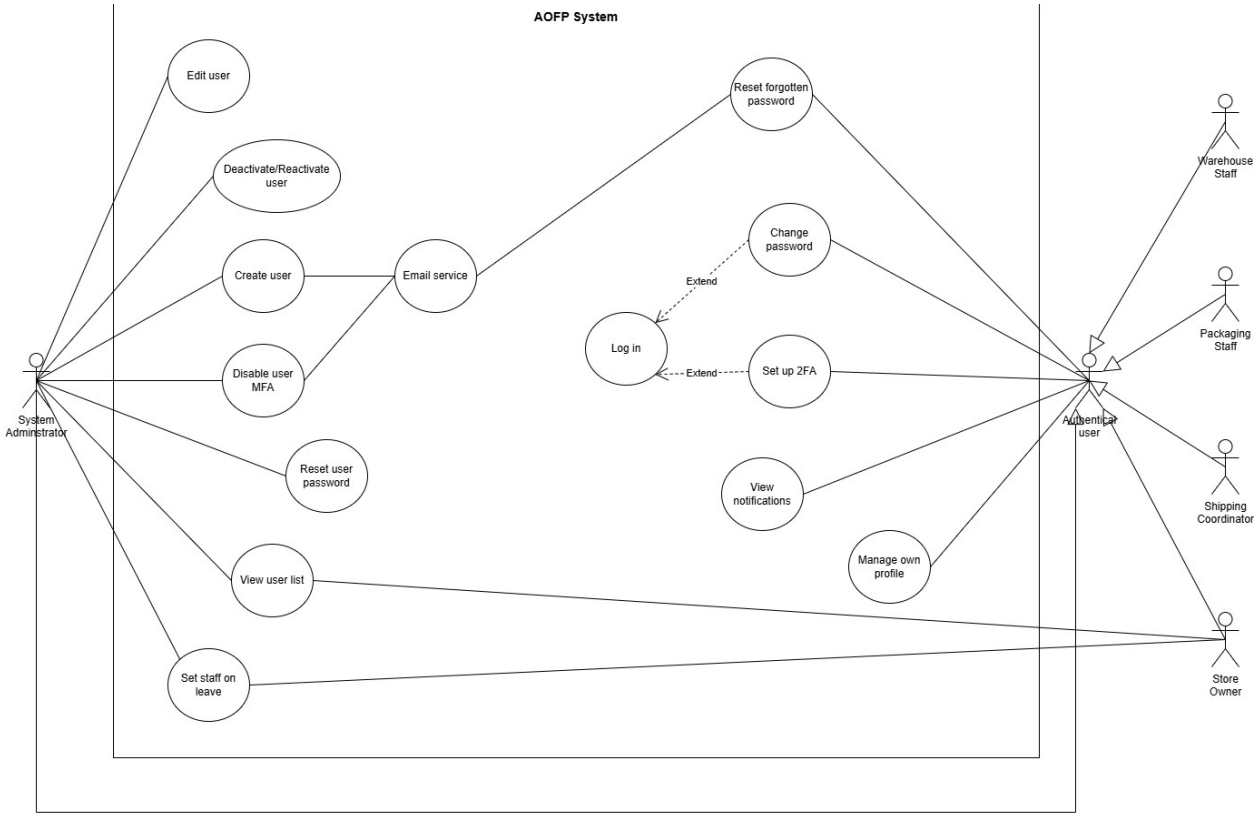


Figure 2. Use case diagram – Account & Security (UC-01 to UC-14)

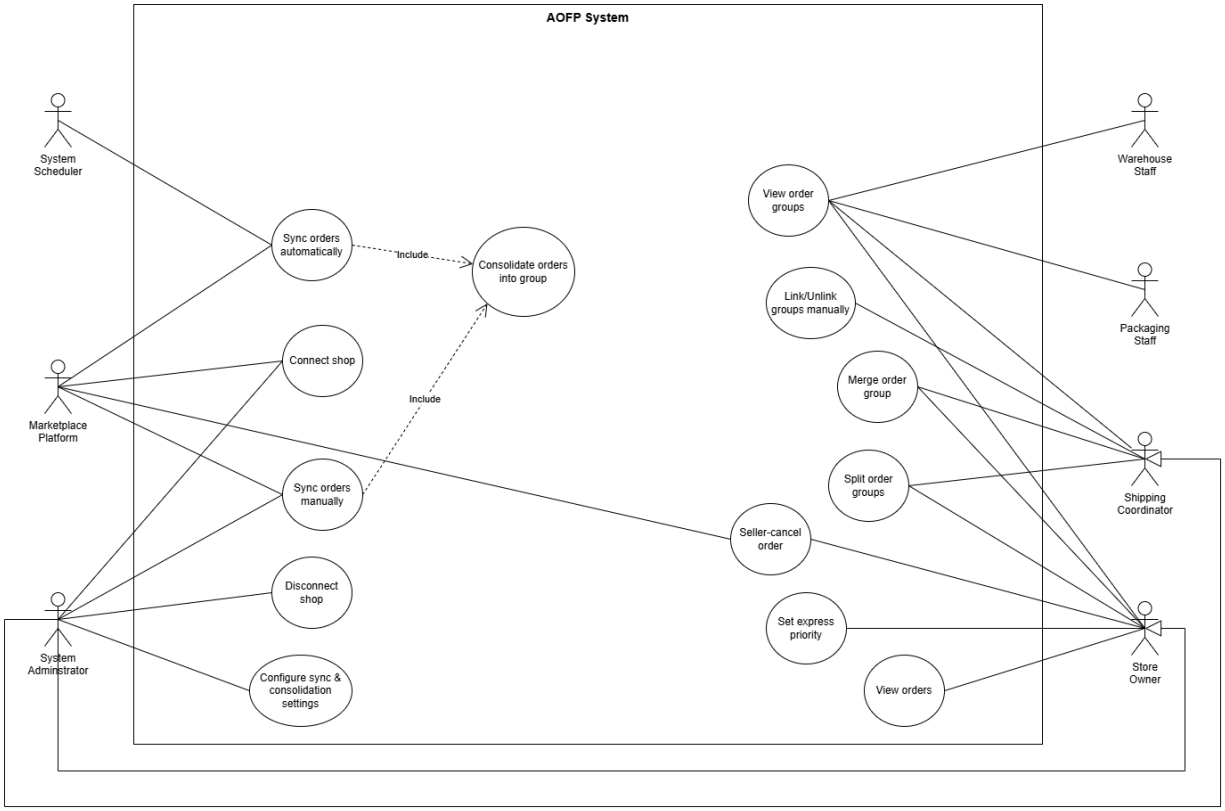


Figure 3. Use case diagram – Marketplace & Orders (UC-15 to UC-27)

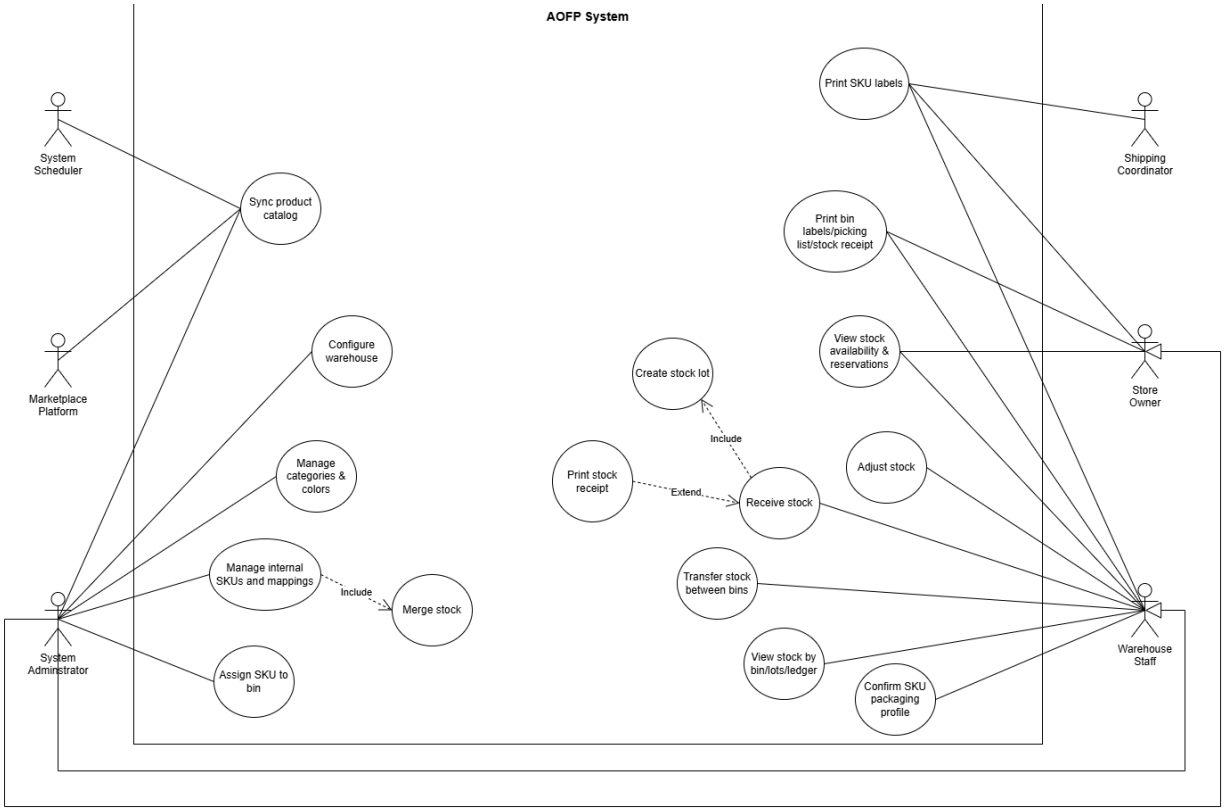


Figure 4. Use case diagram – Warehouse & Inventory (UC-28 to UC-43)

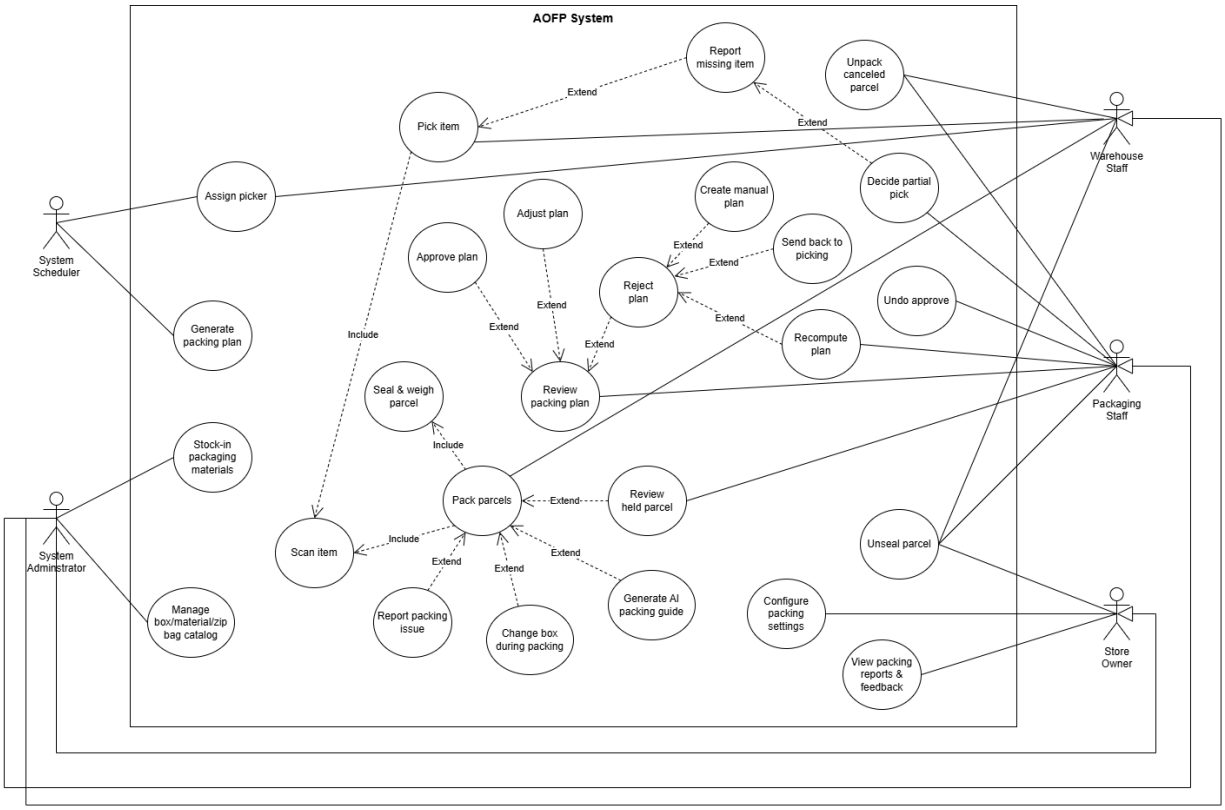


Figure 5. Use case diagram – Picking & Packing (UC-44 to UC-69)

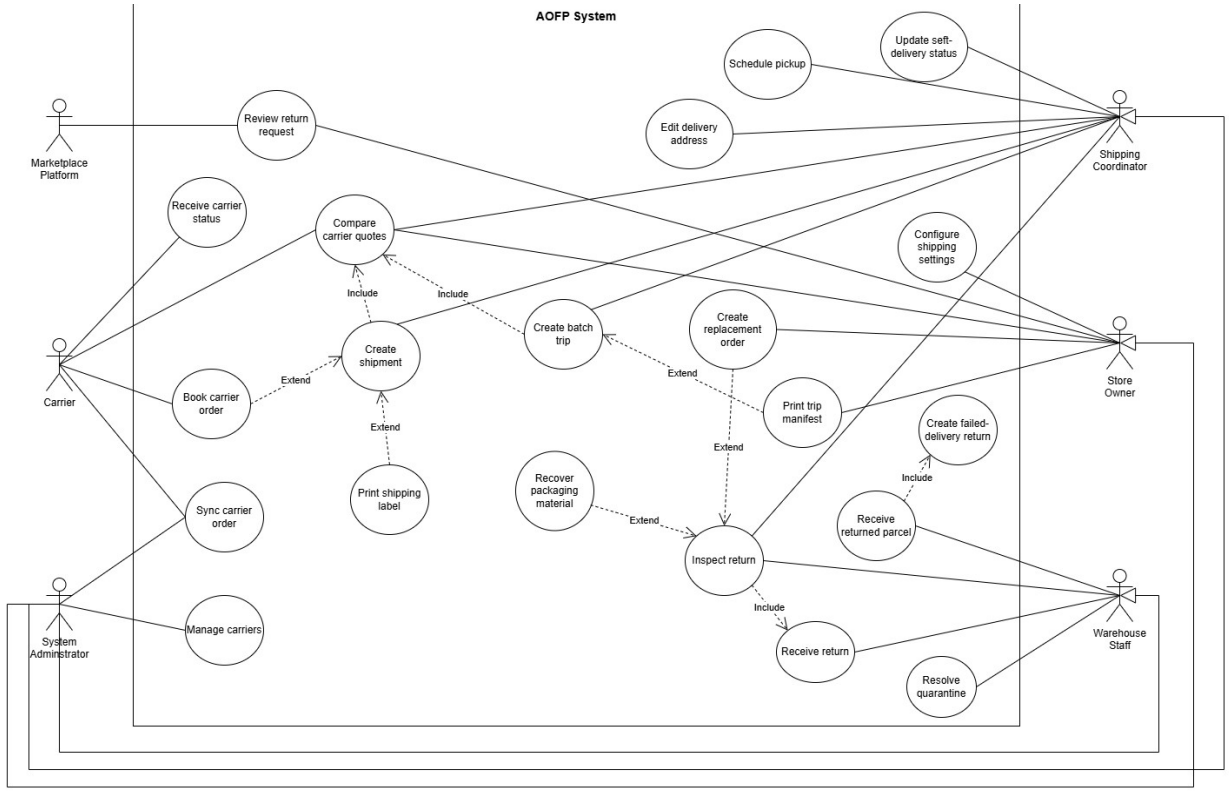


Figure 6. Use case diagram – Shipping & Returns (UC-70 to UC-90)

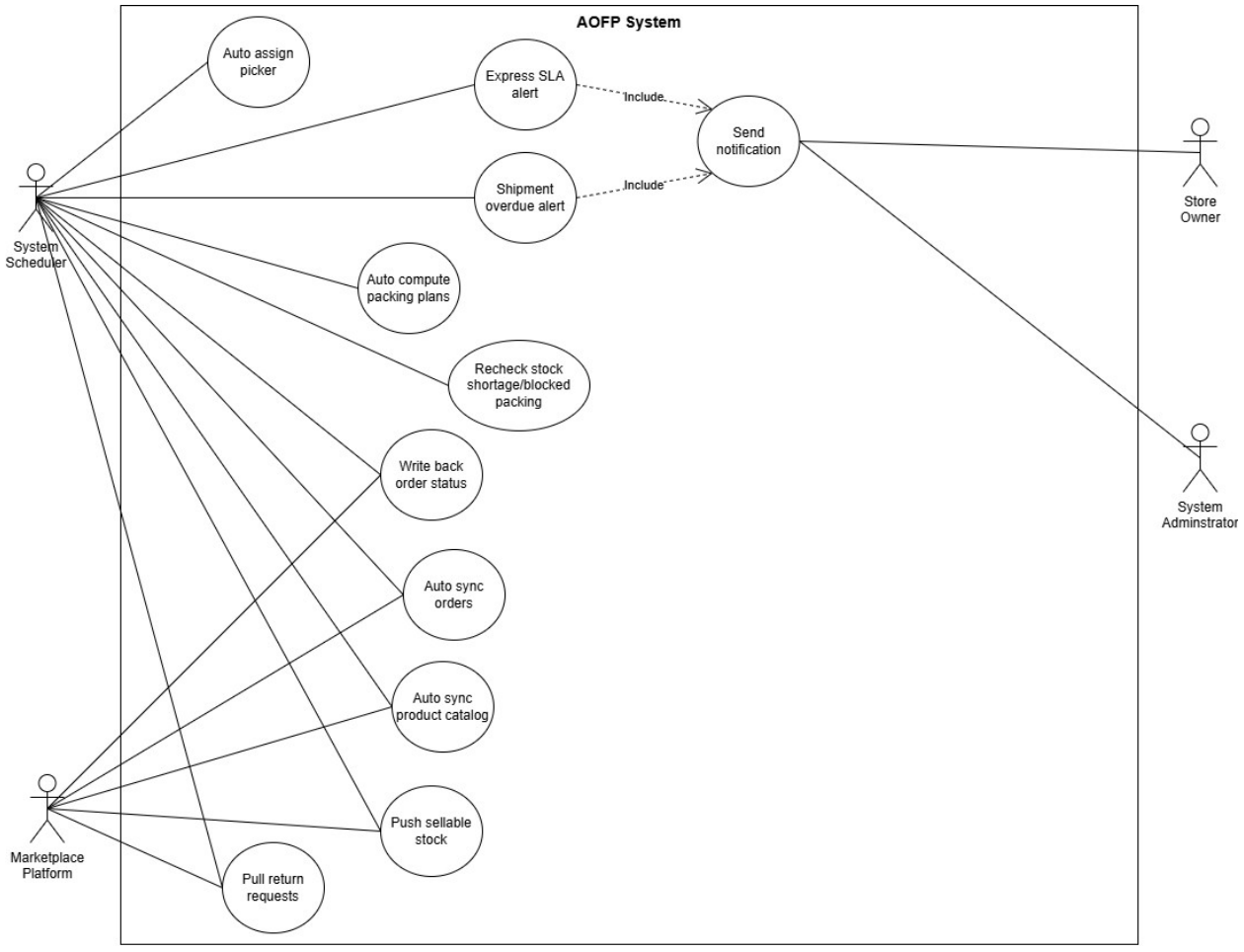


Figure 7. Use case diagram – System Scheduler: background jobs and alerts (UC-91 to UC-101)

2.2.2 Descriptions

ID	Use Case	Actors	Use Case Description
UC-01	Log in	Authenticated user	Log in with e-mail and password or with Google (the e-mail must already exist). After 5 failed attempts the account is locked for 15 minutes. Extended by Change password (temporary password) and Set up 2FA (TOTP or backup code when enabled; a device can be trusted for 30 days).
UC-02	Set up 2FA	Authenticated user	Optional for every role: scan the TOTP secret, confirm a code and receive 10 one-time backup codes. Extends Log in.
UC-03	Change password	Authenticated user	Change the password; forced at the first login with a temporary password. All sessions of the account are signed out. Extends Log in.
UC-04	Reset forgotten password	Authenticated user	Request a reset link by e-mail (Email service) and set a new password before the link expires (30 minutes, single use).
UC-05	Manage own profile	Authenticated user	View and update phone, address and avatar of the own account.
UC-06	View notifications	Authenticated user	View, read and mark all in-app notifications as read; a notification opens the related screen.
UC-07	View user list	System Administrator, Store Owner	List staff accounts with role, status and leave.
UC-08	Create user	System Administrator	Create a staff account with one role; a temporary password is e-mailed (Email service). An e-mail already used by an active or deactivated account is refused.
UC-09	Edit user	System Administrator	Edit name, role, employee code and department of an account.
UC-10	Deactivate / Reactivate user	System Administrator	Deactivate an account (sessions revoked) or reactivate it with its existing data.
UC-11	Reset user password	System Administrator	Issue a new temporary password that must be changed within 72 hours.
UC-12	Disable user MFA	System Administrator	Turn off two-factor authentication of another account; the user is informed in-app and by e-mail (Email service).
UC-13	Set staff on leave	System Administrator, Store Owner	Mark a staff member on leave until a date; staff on leave are not assigned work automatically or manually.
UC-14	Email service	– (used by Create user, Disable user MFA, Reset forgotten password)	Sends account e-mails through SMTP (temporary password, reset link, MFA disabled, account locked). A failed e-mail does not fail the business action.
UC-15	Connect shop	System Administrator, Marketplace Platform	Authorize a Lazada or AURELLE shop through OAuth; tokens are stored encrypted.
UC-16	Disconnect shop	System Administrator	Deactivate a connected shop; it is no longer synchronized.
UC-17	Sync orders automatically	System Scheduler, Marketplace Platform	Every minute import new / changed orders of each shop whose interval (default 5 minutes, 5–1,440) has passed; AURELLE also sends signed order-changed webhooks. Includes Consolidate orders into group.
UC-18	Sync orders manually	System Administrator,	Import the orders of one shop on demand. Includes

ID	Use Case	Actors	Use Case Description
		Marketplace Platform	Consolidate orders into group.
UC-19	Consolidate orders into group	– (included)	Join an open group with the same consolidation key (default platform + phone + address + city) or create a new group; express AURELLE orders always start their own group. A new group is reserved stock and assigned a picker.
UC-20	Configure sync & consolidation settings	System Administrator (Store Owner: view)	Choose imported statuses, consolidation key fields, time window and group statuses that still accept orders; add extra consolidation rules (fields, platforms, time window), optionally drafted by AI from a typed sentence and previewed before saving. Rules never merge orders of different shops; every change is versioned.
UC-21	View order groups	Warehouse Staff, Packaging Staff, Shipping Coordinator, Store Owner	Filter groups by status, priority and platform; search by group id or marketplace order number.
UC-22	Link / Unlink groups manually	Shipping Coordinator	Link groups of the same recipient (e.g. masked Lazada data or different platforms) so they are picked and shipped together; unlink when linked by mistake.
UC-23	Merge order group	Shipping Coordinator, Store Owner	Merge a source group into a target group of the same shop and phone (reason required) before any item is scanned; express groups cannot be merged.
UC-24	Split order groups	Shipping Coordinator, Store Owner	Split an order out of a consolidated group into its own group (reason required) before any item is scanned.
UC-25	Seller-cancel order	Store Owner, Marketplace Platform	Cancel an AURELLE order that cannot be fulfilled until it is handed to the carrier; the reason is written back to AURELLE and a paid order is refunded.
UC-26	Set express priority	Store Owner	Mark a group as express with a packaging deadline in business hours; the system warns before the deadline.
UC-27	View orders	Store Owner	List and view orders of all connected channels with items, recipient and status.
UC-28	Sync product catalog	System Scheduler, Marketplace Platform, System Administrator	Import the marketplace catalog hourly, fully at 03:00 every day or on demand; seller SKUs that follow the internal code rule are mapped or created automatically.
UC-29	Configure warehouse	System Administrator	Create and edit warehouses, zones, racks and bins with capacity; deactivate, or delete when there is no stock history.
UC-30	Manage categories & colors	System Administrator	Manage product categories (with size scales) and colours used in internal SKU codes.
UC-31	Manage internal SKUs and mappings	System Administrator	Manage internal SKUs (CATEGORY-MODEL-COLOUR-SIZE) and map marketplace seller SKUs to them. Includes Merge stock.
UC-32	Merge stock	– (included)	Pool the stock and reservations of mapped seller SKUs under the internal SKU so all channels share it.
UC-33	Assign SKU to bin	System Administrator	Assign a SKU to a bin with an optional initial quantity; a second bin of the same warehouse needs explicit confirmation.
UC-34	Receive stock	Warehouse Staff	Create a stock receipt (lot code, received date, supplier, lines); it is refused when a bin capacity

ID	Use Case	Actors	Use Case Description
			would be exceeded unless forced. Includes Create stock lot; extended by Print stock receipt.
UC-35	Create stock lot	– (included)	Each receipt line becomes a lot dated by its received date; picking consumes the oldest lot first (FIFO).
UC-36	Print stock receipt	Warehouse Staff	Download the stock receipt as PDF. Extends Receive stock.
UC-37	Adjust stock	Warehouse Staff	Stocktake: enter the counted quantity and a reason; the difference is written to the inventory ledger.
UC-38	Transfer stock between bins	Warehouse Staff	Move a quantity to another bin of the same warehouse; lots keep their received date.
UC-39	View stock by bin / lots / ledger	Warehouse Staff	View bin stock, lots and the movement history of a SKU in a bin.
UC-40	View stock availability & reservations	Warehouse Staff, Store Owner	View on-hand, reserved and available quantity per SKU and the reservation of an order group.
UC-41	Confirm SKU packaging profile	Warehouse Staff	Confirm package dimensions, weight, fragility, item category and folding of a SKU; packing waits until every SKU of a group is confirmed.
UC-42	Print SKU labels	Warehouse Staff, Packaging Staff, Store Owner	Print SKU labels as PDF (1–50 copies per SKU).
UC-43	Print bin labels / picking list / stock receipt	Warehouse Staff, Store Owner	Print bin labels of a zone or bins, the picking list of a group and stock receipts as PDF.
UC-44	Assign picker	System Scheduler, Warehouse Staff	Automatic: the least-busy Warehouse Staff not on leave when a group is created; manual reassignment from the picking screen.
UC-45	Pick item	Warehouse Staff	Pick the items of an assigned group bin by bin and confirm picking completed when every SKU is fully picked. Includes Scan item; extended by Report missing item.
UC-46	Scan item	– (included by Pick item and Pack parcels)	Check a scanned (or manually entered) SKU against the group or parcel; a repeated scan with the same client event id is not counted twice.
UC-47	Report missing item	Warehouse Staff	Report a SKU that cannot be found (missing quantity, note); the group goes to partial_needs_review. Extends Pick item.
UC-48	Decide partial pick	Packaging Staff	Ship the available items (shortage allocated to the oldest orders, customers informed and not charged) or start a new pick round. Extends Report missing item.
UC-49	Generate packing plan	System Scheduler	Every 10 seconds compute a 3D packing plan for each order of a picked group (BRKGA, optional CP-SAT proof); a group with an unconfirmed SKU profile is blocked.
UC-50	Review packing plan	Packaging Staff	View the 3D plan of each order with boxes, materials, cost and box comparison. Extended by Approve plan, Adjust plan and Reject plan.
UC-51	Approve plan	Packaging Staff	Approve a plan in which every item is placed; the group moves to approved_for_packing. Extends Review packing plan.
UC-52	Adjust plan	Packaging Staff	Change the box of a parcel or move an item to another parcel of the same order; every change is

ID	Use Case	Actors	Use Case Description
			validated. Extends Review packing plan.
UC-53	Reject plan	Packaging Staff	Reject with a reason code ("Other" needs a note); Admin and Store Owner must handle it within 2 working hours. Extends Review packing plan; extended by Create manual plan, Send back to picking and Recompute plan.
UC-54	Create manual plan	Packaging Staff	Enter the actual parcels (boxes and items); the manual plan must be approved by another person.
UC-55	Send back to picking	Packaging Staff	Declare the items to re-pick; the group returns to picking.
UC-56	Recompute plan	Packaging Staff	Compute a new plan excluding boxes or preferring a cheaper plan.
UC-57	Undo approve	Packaging Staff	Before the first parcel is sealed, return an approved plan to review (group back to pending_approval, scans cleared).
UC-58	Pack parcels	Packaging Staff, Warehouse Staff	Take the box shown for each parcel (reused shelf first, new box for fragile items) and scan the items into it. Includes Scan item and Seal & weigh parcel; extended by Review held parcel, Report packing issue, Change box during packing and Generate AI packing guide.
UC-59	Seal & weigh parcel	– (included)	Seal the parcel, deduct box and materials and enter the real weight; a weight outside the tolerance (default 20%) puts the parcel on hold.
UC-60	Review held parcel	Packaging Staff	Accept, reweigh or reopen a held parcel; the reviewer must be a different person from the one who sealed it.
UC-61	Report packing issue	Packaging Staff, Warehouse Staff	Report damaged, missing or wrong items and replace them from the shelf.
UC-62	Change box during packing	Packaging Staff, Warehouse Staff	Use another box for a parcel during packing (e.g. a new box instead of a reused one); a damaged box is written off.
UC-63	Generate AI packing guide	Packaging Staff, Warehouse Staff	Generate the step-by-step wording of the packing guide (Groq; template text when unavailable) and rate it.
UC-64	Unseal parcel	Packaging Staff, Warehouse Staff, Store Owner	Reopen a sealed parcel before it is shipped.
UC-65	Unpack canceled parcel	Packaging Staff, Warehouse Staff	Unpack a parcel of a cancelled order: items return to the bins they were picked from, a reusable box and fillers are recovered.
UC-66	Configure packing settings	Store Owner	Set weight tolerance, fragile cushion, preferred plan, parcel limit and scan requirement.
UC-67	View packing reports & feedback	Store Owner	View packing reports (plans, rejections, packaging cost, incidents, per staff) and rejection feedback.
UC-68	Stock-in packaging materials	System Administrator (also Warehouse Staff)	Record purchases and internal use of boxes, zip bags and fillers; an alert is sent when stock nears or falls below the reorder level.
UC-69	Manage box/material/zip bag catalog	System Administrator	Manage boxes, filling materials and zip bags with dimensions, cost and reorder level.
UC-70	Compare carrier quotes	Shipping Coordinator, Store Owner, Carrier	Quote each carrier service with fee, chargeable weight and expected delivery time (details in 3.7.1).

ID	Use Case	Actors	Use Case Description
UC-71	Create shipment	Shipping Coordinator	Create a shipment for a packed group with the chosen carrier and service. Includes Compare carrier quotes; extended by Book carrier order and Print shipping label.
UC-72	Book carrier order	Carrier	One carrier (GHN) order per parcel with the recipient of that parcel's order; parcels without tracking code can be re-booked.
UC-73	Print shipping label	Shipping Coordinator (also Warehouse / Packaging Staff)	Download the carrier shipping label (A5 PDF). Extends Create shipment.
UC-74	Create batch trip	Shipping Coordinator	Ship several packed groups of the same recipient as one trip. Includes Compare carrier quotes; extended by Print trip manifest.
UC-75	Print trip manifest	Store Owner, Shipping Coordinator	Download the manifest of a trip as PDF. Extends Create batch trip.
UC-76	Schedule pickup	Shipping Coordinator	Record or change the time the carrier picks up the shipment (stored in OptiPack; no carrier API call).
UC-77	Edit delivery address	Shipping Coordinator	Correct street, ward, province or phone of a parcel rejected by the carrier, then re-book it.
UC-78	Update self-delivery status	Shipping Coordinator	For an express AURELLE group delivered by a shop shipper: record delivered or failed delivery.
UC-79	Receive carrier status	Carrier	GHN status webhooks are mapped to the shipment status; duplicate events are ignored.
UC-80	Sync carrier order	Carrier, System Administrator (also Shipping Coordinator)	Pull the latest status of a carrier order on demand.
UC-81	Manage carriers	System Administrator	Manage carriers and services (volumetric divisor, sample rate tables for carriers without API).
UC-82	Configure shipping settings	Store Owner	Sender name, phone and address printed on labels and the carrier selection strategy (cheapest, fastest or fixed).
UC-83	Review return request	Marketplace Platform, Store Owner	A return request is pulled from AURELLE; the Store Owner approves, rejects (reason required) or cancels it; the creator of a ticket cannot approve it.
UC-84	Receive returned parcel	Warehouse Staff	Scan a parcel returned by the carrier. Includes Create failed-delivery return.
UC-85	Create failed-delivery return	– (included)	The shipment becomes returned_to_warehouse, the group becomes returned and a failed-delivery return ticket is opened for inspection.
UC-86	Inspect return	Warehouse Staff	Decide restock, quarantine or discard for each returned SKU; restocked items go back to a bin. Includes Receive return; extended by Recover packaging material and Create replacement order.
UC-87	Receive return	Warehouse Staff	Mark the goods sent back by the customer as received.
UC-88	Recover packaging material	Warehouse Staff	Return reusable boxes and materials of the returned parcel to stock. Extends Inspect return.
UC-89	Create replacement order	Store Owner	For an exchange, create the EXC-… replacement order (price 0) and a new order group; retry when it failed. Extends Inspect return.
UC-90	Resolve quarantine	Warehouse Staff	Restock or discard quarantined items later.
UC-91	Auto sync orders	System Scheduler, Marketplace Platform	Every minute, for each shop whose interval has passed (default 5 minutes).

ID	Use Case	Actors	Use Case Description
UC-92	Auto sync product catalog	System Scheduler, Marketplace Platform	Hourly catalog sync and full sync at 03:00 (Asia/Ho_Chi_Minh).
UC-93	Auto assign picker	System Scheduler	Least-busy Warehouse Staff not on leave, when the group is created; a backfill job every 5 minutes resumes groups stuck before picking.
UC-94	Auto compute packing plans	System Scheduler	Every 10 seconds for picked groups without an active plan.
UC-95	Recheck stock shortage / blocked packing	System Scheduler	Re-reserve stock of groups with shortage after stock-in; every minute retry groups blocked by unmeasured SKUs.
UC-96	Write back order status	System Scheduler, Marketplace Platform	Every 15 seconds send queued order / return status changes to AURELLE (up to 8 attempts with back-off).
UC-97	Push sellable stock	System Scheduler, Marketplace Platform	Every minute push available stock of mapped SKUs to AURELLE.
UC-98	Pull return requests	System Scheduler, Marketplace Platform	Every 2 minutes pull AURELLE return requests and cancellations not received by webhook.
UC-99	Express SLA alert	System Scheduler	Every minute warn before the packing / delivery deadline of an express group and flag overdue groups. Includes Send notification.
UC-100	Shipment overdue alert	System Scheduler	Every 15 minutes mark overdue shipments. Includes Send notification.
UC-101	Send notification	Store Owner, System Administrator (recipients)	Create the in-app notification (and e-mail by role) for the alert.
UC-102	Register and log in (customer)	Customer	Create an AURELLE account and log in (AURELLE website).
UC-103	Browse products and manage cart	Customer	Browse categories and products, add variants to the cart, update quantities.
UC-104	Place order	Customer	Check out with a shipping address (province + ward) and payment method (COD or bank transfer).
UC-105	Track and cancel order	Customer	Follow the 7-step order progress and delivery events; cancel while the order is pending and not packed.
UC-106	Request return or exchange	Customer	Request return & refund, refund only or exchange within 15 days of delivery; cancel a pending request.
UC-107	Manage storefront catalog	AURELLE Web Admin	Manage categories, products, variants (SKU by OptiPack rule) and images.
UC-108	Manage developer apps	AURELLE Web Admin	Create Open Platform apps, rotate the secret, revoke / reactivate keys.

3. Functional Requirements

3.1 System Functional Overview

OptiPack AI has two web front ends served by one back end: the OptiPack staff portal (React) and the AURELLE website with its customer, web admin and developer pages (Next.js). The AURELLE Open API used as the second marketplace runs as a separate simulator process. Staff roles see only the screens allowed for them (section 3.1.3). Main business workflows WF-01 to WF-06 are described with activity diagrams in Report 4 (SDD), section "Main Business Workflows".

3.1.1 Screens Flow

The screens of the OptiPack staff portal and the AURELLE website, with the roles that use them, are listed in section 3.1.2 Screen Descriptions and section 3.1.3 Screen Authorization.

3.1.2 Screen Descriptions

#	Feature	Screen	Description
1	Common	Login	E-mail / password and Google login, two-factor step when enabled for the account, link to forgot password.
2	Common	Forgot / Reset Password	Request a reset link and set a new password from the e-mailed link.
3	Common	Change Password	Forced change of the temporary password on first login.
4	Common	Profile & Settings	Personal information, password change, optional two-factor setup.
5	Common	Notifications (top bar)	Unread counter and list; clicking a notification opens the related screen.
6	Dashboard	Dashboard Overview	Home screen of Store Owner / Admin: every widget reads live figures from the back end (GET /dashboard/summary: orders and revenue by channel, fulfilment, shipping and returns, packaging cost and savings).
7	Orders	Omnichannel Orders / Order Detail	Orders of all channels with filters; detail shows items, recipient and status history.
8	Orders	Groups & Express	Order groups with status, priority, packer, shortage and blocked flags; detail with orders, link, priority.
9	Orders	Order Sync Settings	Consolidation key fields, imported statuses, time window, change history.
10	Warehouse	Warehouse Picking	Assigned groups, picking list per SKU (bin, oldest lot, other bins), scan, report missing, confirm picked.
11	Warehouse	Warehouse Inventory	Bin stock, assign SKU to bin, add / adjust / transfer stock, stocktake, stock receipts and lots.
12	Warehouse	Admin Warehouse	Warehouses, zones, racks, bins, internal SKUs, categories, colours, SKU mappings, product master.
13	Packing	Packing Station	Plans waiting for approval, 3D viewer, approve / reject, packing session (scan, seal, weigh), held parcels, issues.
14	Packing	Admin Packing Plans	All packing plans with status, proof label and recompute.
15	Packing	Packaging Rules / Packaging Materials	Boxes, bags, filling materials, rules, purchases, internal use and savings.
16	Shipping	Shipping & Fulfillment	Packed groups, quotes, create shipment / trip, carrier parcels and tracking, failed delivery, labels.
17	Returns	Returns	Return tickets (customer requests and failed deliveries): approve / reject, receive, inspect, quarantine, replacement.
18	People	Staff Directory	Staff list, roles, leave status.
19	Admin	Admin Home / Roles	User accounts, roles, two-factor reset, reactivation.
20	Admin	Marketplace Connections	Connect / disconnect Lazada and AURELLE shops, shop status.
21	Admin	AI & Packing Settings / Analytics	Packing settings, packing reports and feedback.
22	AURELLE	Home / Shop / Product Detail	Customer catalog, product with variants and sellable stock.
23	AURELLE	Cart / Checkout	Cart and checkout with new two-level address (province, ward) and payment method.
24	AURELLE	My Orders / Order Detail	Order list and detail with 7-step progress bar, events, cancel and return actions.
25	AURELLE	My Returns / Return Detail	Return and exchange requests with status and exchange shipment progress.
26	AURELLE	Customer Login / Register	Customer account access; Admin web accounts are redirected to the admin page.
27	AURELLE	Web Admin (Products, Categories)	Storefront catalog management with SKU built from OptiPack categories, colours and sizes.
28	AURELLE	Developer Portal	Open Platform apps: app key / secret, rotate, revoke.

3.1.3 Screen Authorization

X = screen available. Back-end endpoints enforce the same roles (role-based guard on every request).

Screen	Admin	Store Owner	Warehouse Staff	Packaging Staff	Shipping Coord.
Dashboard Overview	X	X			
Omnichannel Orders / Order Detail	X	X			
Groups & Express	X	X			
Order Sync Settings (edit)	X	view			
Returns	X	X			
Warehouse Picking	X		X		
Warehouse Inventory	X		X		
Admin Warehouse (structure, SKUs)	X				
Packing Station	X			X	
Packaging Materials	X	X	X	X	
Packaging Rules	X	X			
Shipping & Fulfillment	X				X
Staff Directory	X	X			
Analytics	X	X			
Admin Home / Roles / Marketplace / Packing Plans / AI	X				
Profile & Settings, Notifications	X	X	X	X	X
Action History (own categories)	X	X	X	X	X
Export history CSV	X	X			

3.1.4 Non-Screen Functions

#	Feature	System Function	Description
1	Orders	Marketplace order auto-sync	Every minute: for each connected Lazada / AURELLE shop whose sync interval has passed (default 5 minutes, configurable 5–1,440), import new / changed orders page by page, consolidate and create groups.
2	Orders	Marketplace webhooks	Receive AURELLE webhooks (order changed, product updated, return requested / cancelled), verify HMAC signature, ignore duplicates.
3	Orders	Order-group backfill	Every 5 minutes: create missing groups and resume groups stuck in awaiting_packaging.
4	Orders	Express SLA check	Every minute: warn before the packing and delivery deadlines of an express group and flag overdue groups (each warning is sent once).
5	Catalog	Product master sync	Hourly catalog sync and daily full sync at 03:00 (Asia/Ho_Chi_Minh); auto-map and auto-create internal SKUs that follow the code rule.
6	Packing	Packing-plan auto compute	Every 10 seconds: compute plans for picked groups (BRKGA solver, optional CP-SAT proof service).
7	Packing	Packing-blocked recheck	Every minute: retry groups blocked by unmeasured SKUs after the SKU is confirmed.
8	Packing	Rejected plan reminder	Every 15 minutes: remind once when a rejected plan passes its 2-working-hour deadline.
9	Shipping	Carrier webhook	Receive GHN status callbacks, map to shipment status, ignore

#	Feature	System Function	Description
			duplicates.
10	Shipping	Shipment SLA scan	Every 15 minutes: mark overdue deliveries.
11	Write-back	Order / return status write-back	Every 15 seconds: send queued status changes to AURELLE, retry with back-off (up to 8 attempts).
12	Write-back	Stock push	Every minute: push available stock of mapped SKUs to AURELLE.
13	Returns	Marketplace returns sync	Every 2 minutes: pull AURELLE return requests and cancellations not received by webhook.
14	Common	Notification and e-mail	Create in-app notifications by role / user; send account e-mails through SMTP.

3.1.5 Entity Relationship Diagram

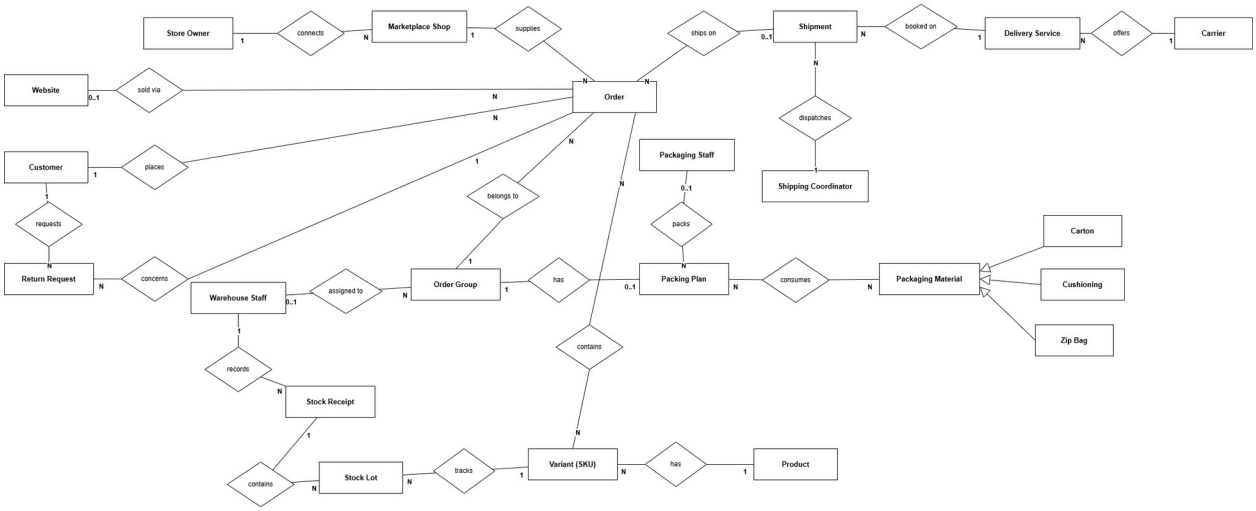


Figure 8. Logical entity relationship diagram

Entities Description

#	Entity	Description
1	Store Owner	Business owner account (users collection, role Store Owner). The marketplace shops belong to the owner's business; the OAuth connection itself is performed by the Admin.
2	Marketplace Shop	A connected Lazada or AURELLE shop with encrypted tokens and connection status (marketplace_shops); supplies orders.
3	Website	The AURELLE website through which orders can be sold; an order is sold via at most one website (orders with platform AURELLE; website data in the storefront_* collections).
4	Customer	Buyer who places orders and requests returns (customers collection for AURELLE accounts; Lazada recipients are stored on the order).
5	Order	Marketplace order with recipient, items, status and consolidation key (orders); contains SKU variants.
6	Return Request	Return, refund-only, exchange or failed-delivery ticket concerning one order, with inspection results (return_requests).
7	Order Group	Unit of fulfilment: one or more orders of the same recipient that are picked, packed and shipped together (order_groups).
8	Warehouse Staff	Staff account (users, role Warehouse Staff) assigned to order groups as picker and recording stock receipts.
9	Packaging Staff	Staff account (users, role Packaging Staff) assigned as packer of packing plans.
10	Shipping Coordinator	Staff account (users, role Shipping Coordinator) who dispatches shipments.

#	Entity	Description
11	Packing Plan	3D packing plan of an order group: parcels, boxes, placements, scans and weighings (packing_plans); a group has at most one active plan.
12	Packaging Material	Generalization of Carton, Cushioning and Zip Bag with new / reused stock, reorder level and movements (packaging_materials, packaging_bags, packaging_movements); consumed by packing plans.
13	Carton / Cushioning / Zip Bag	Subtypes of Packaging Material: boxes (inner and outer dimensions), filling materials, and zip bags that wrap an item.
14	Shipment	Outbound shipment with carrier parcels and status history (shipments, shipment_events); dispatched by a Shipping Coordinator and booked on a delivery service.
15	Delivery Service	A service of a carrier (e.g. GHN standard) with its volumetric divisor (shipping_carriers).
16	Carrier	Shipping company offering delivery services (GHN live API; other carriers with sample rate tables).
17	Stock Receipt	Goods receipt recorded by Warehouse Staff with lot code, received date, supplier and lines (stock_receipts).
18	Stock Lot	Quantity of one SKU received on a date, consumed oldest first (stock_lots).
19	Variant (SKU)	Sellable variant identified by its SKU (internal master_skus mapped to marketplace seller SKUs; product_master holds its packaging profile).
20	Product	Product that groups its variants (marketplace catalog products; storefront_products on AURELLE).

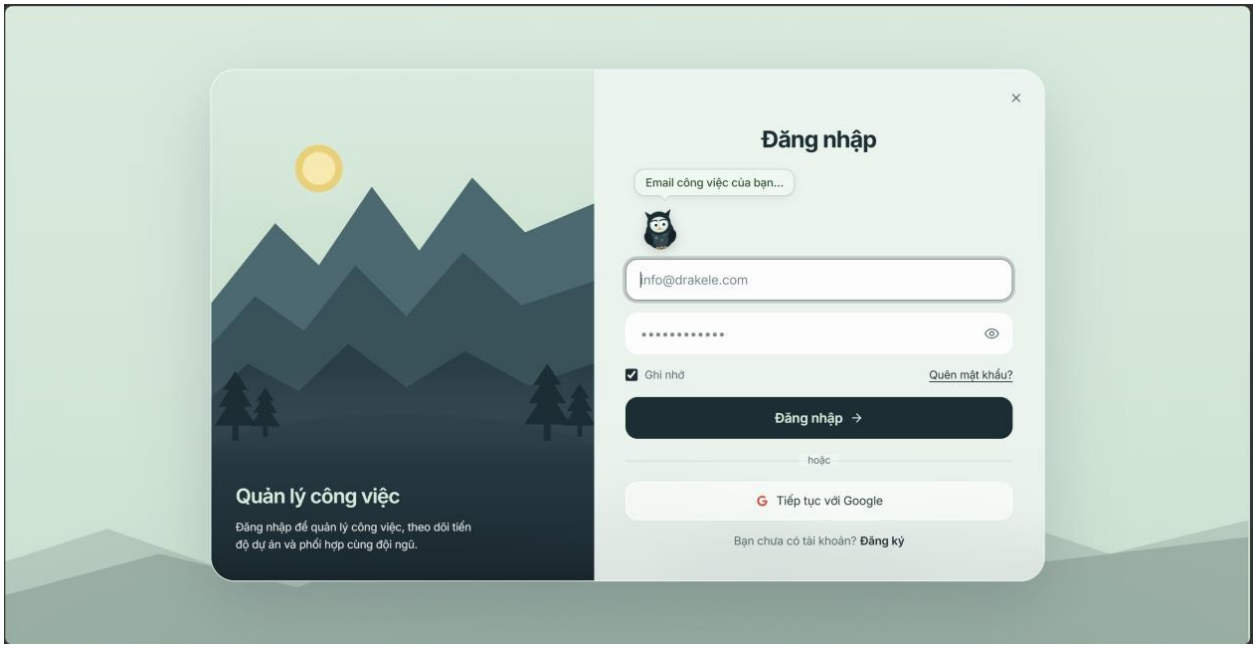
3.2 Authentication and Account Management

3.2.1 Log in

Function trigger: Login screen; Google button; /login.

Function description: All staff roles. Authenticates the user and opens the home screen of the role.

Screen layout:



Function details:

- E-mail and password are required; a wrong password increases the failed-attempt counter. After 5 consecutive failures the account is locked for 15 minutes.

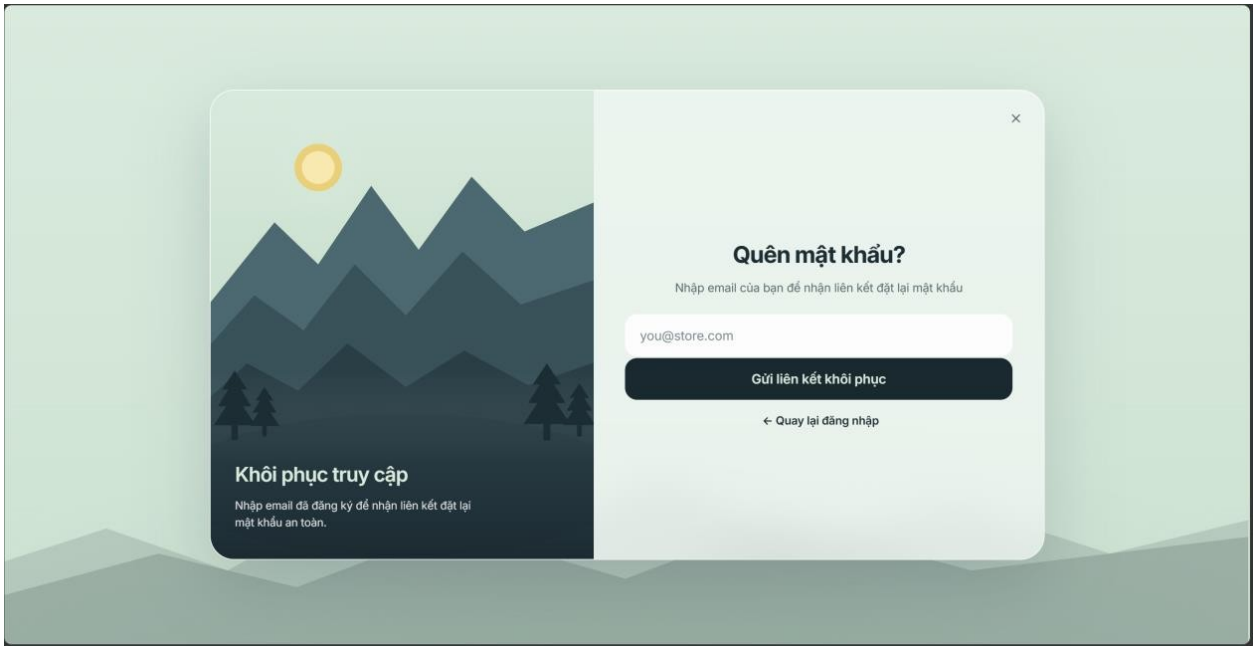
- Google login succeeds only when the verified Google e-mail already belongs to an active account.
- A user who has enabled two-factor authentication must enter a TOTP code or a backup code unless the device is trusted (30 days); two-factor authentication is optional for every role.
- An account created by Admin has a temporary password that must be changed within 72 hours; otherwise the account is locked until Admin reactivates it.
- Access token lifetime 24 hours, refresh token 30 days with rotation; reuse of an old refresh token revokes the session.

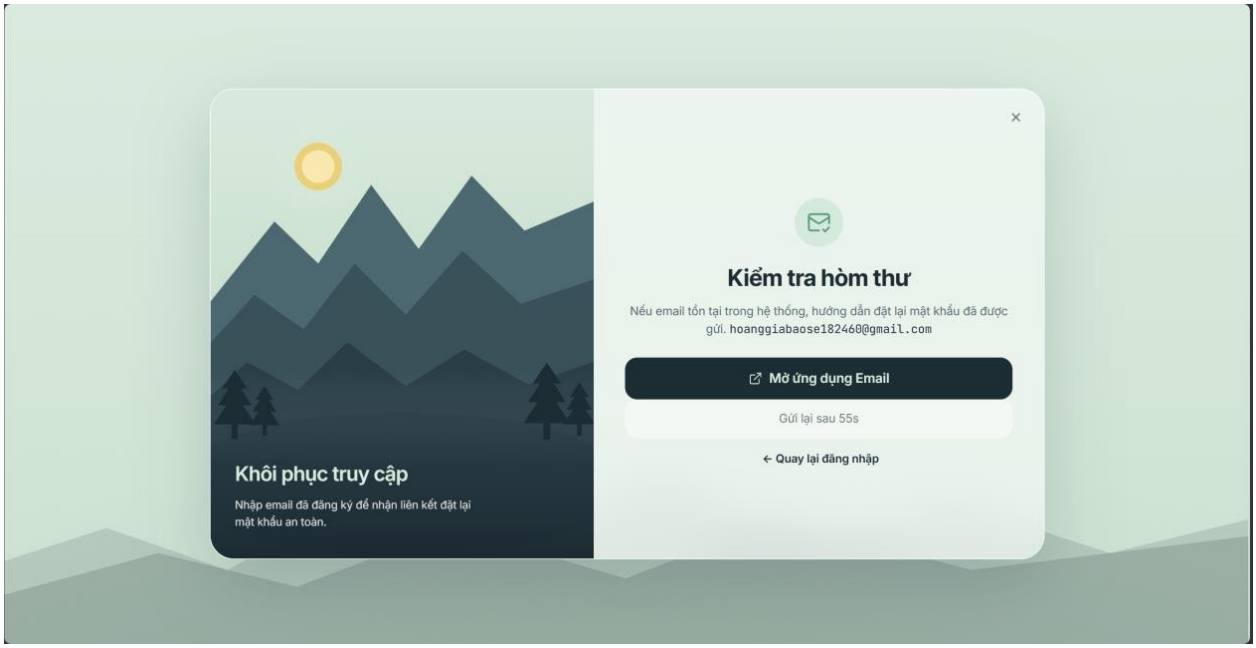
3.2.2 Reset and change password

Function trigger: Forgot Password link; Change Password screen.

Function description: All staff roles. Recovers access or replaces the password.

Screen layout:





Function details:

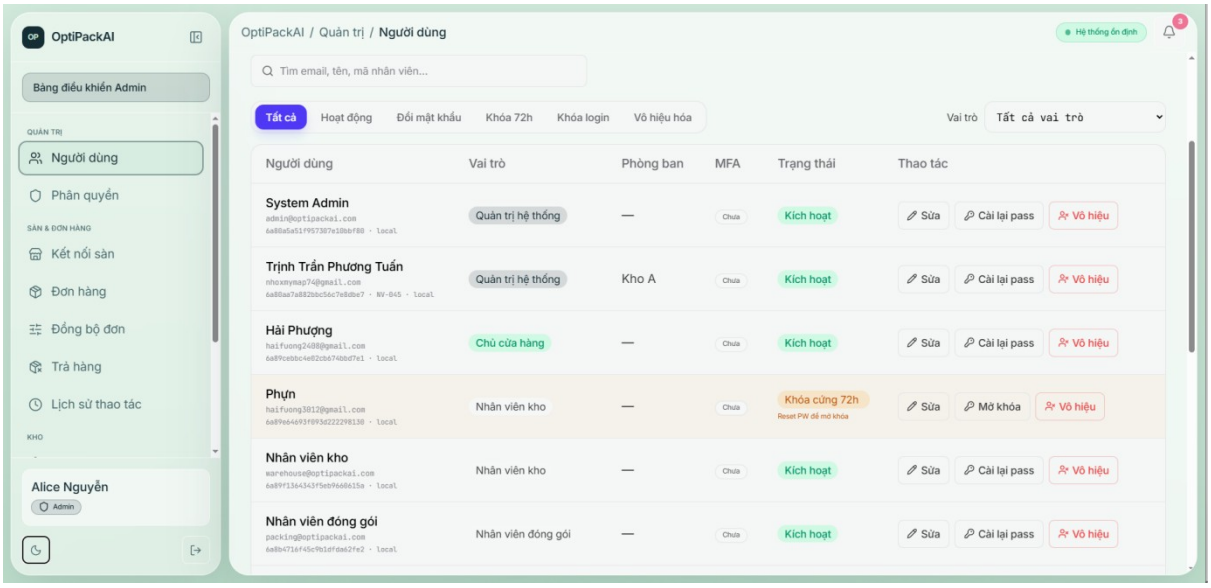
- The reset e-mail always shows the same generic message whether the e-mail exists or not.
- The reset link expires after 30 minutes and can be used once.
- New password: at least 8 characters with upper case, lower case, digit and special character.
- Changing or resetting the password signs out all sessions of the account.

3.2.3 Manage user accounts and leave

Function trigger: Admin Home / Roles; Staff Directory.

Function description: Admin manages accounts; Store Owner views staff and sets leave.

Screen layout:



Function details:

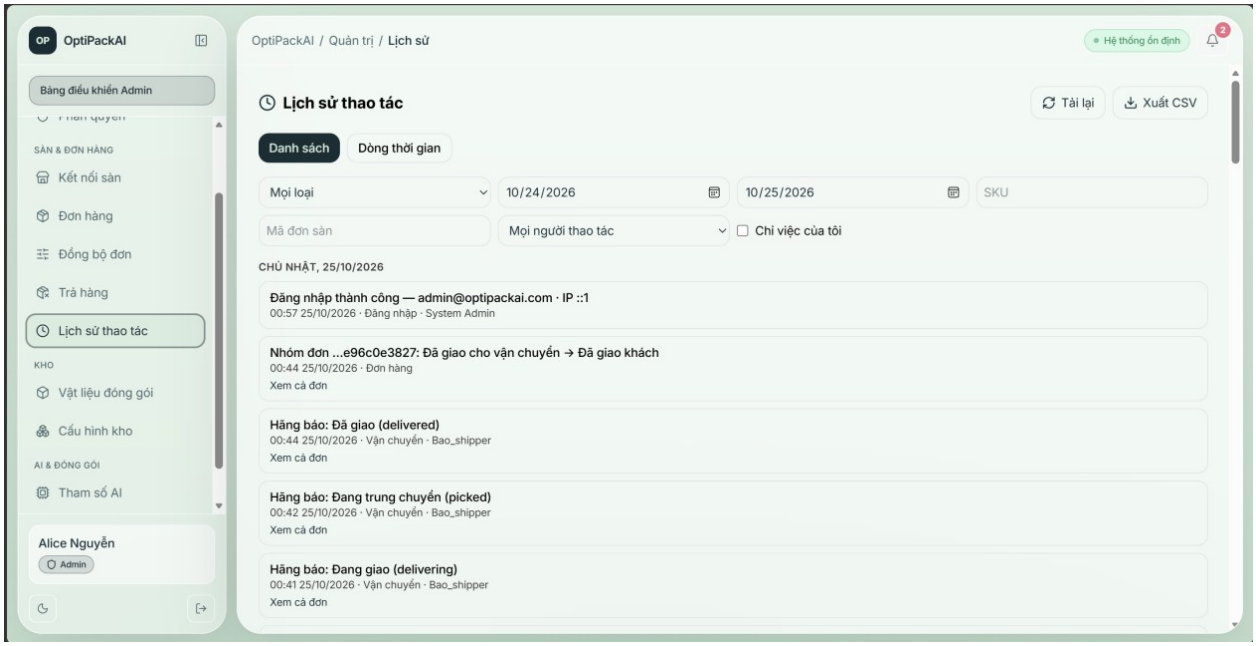
- Admin creates the account and the system e-mails a temporary password.
- Deactivating an account revokes its sessions; reactivation clears lock state.
- Staff on leave (until a date) are skipped by auto-assignment and cannot be assigned manually.

3.2.4 Action history

Function trigger: History screen; Timeline tab of an order group.

Function description: All roles; each role sees only its categories. Admin and Store Owner export CSV (max 5,000 rows).

Screen layout:



Function details:

- Categories: orders, picking, packing, shipping, returns, warehouse, packaging materials, login (Admin only).
- Default period 30 days; filter by date, SKU, order code, actor, "my actions".

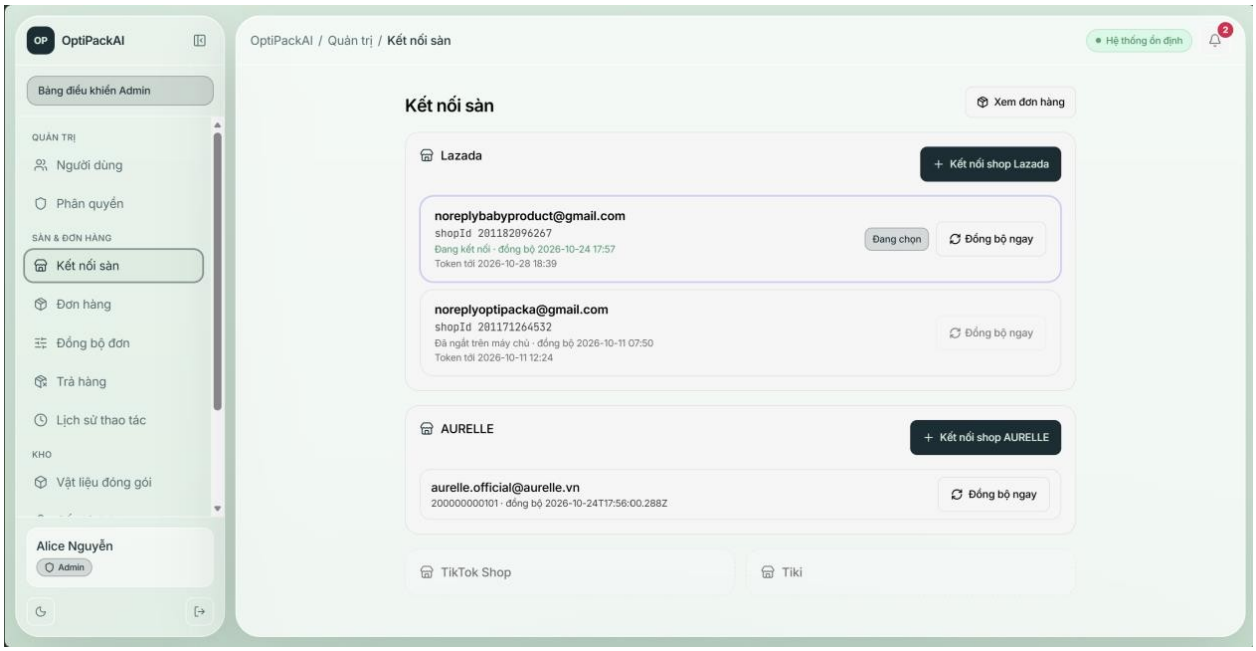
3.3 Marketplace Integration and Order Intake

3.3.1 Connect / disconnect shop

Function trigger: Admin → Marketplace Connections.

Function description: Admin authorizes a Lazada or AURELLE shop through OAuth.

Screen layout:



Function details:

- Tokens are encrypted (AES-256-GCM) before storage and refreshed automatically.

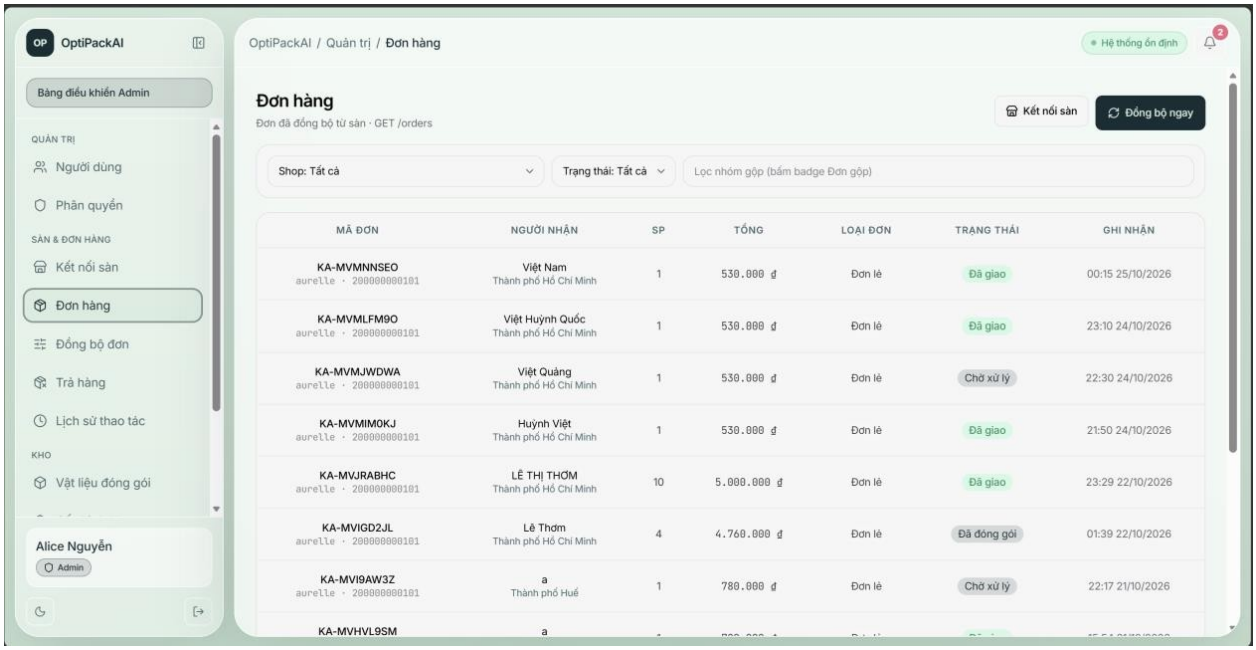
- A disconnected shop is no longer synchronized; reconnecting reactivates it.

3.3.2 Synchronize and consolidate orders

Function trigger: Automatic (scheduler every minute, each shop at its interval, default 5 minutes), webhook (AURELLE), or "Sync" button (Admin).

Function description: System imports orders and builds order groups.

Screen layout:



Function details:

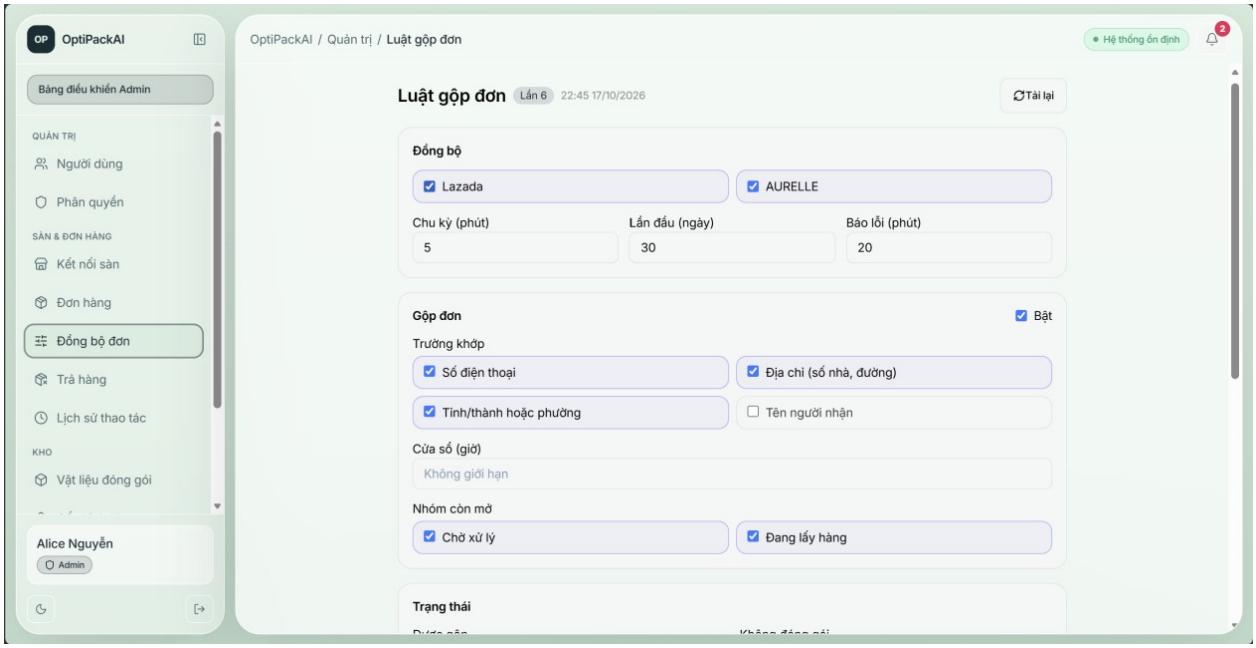
- Orders with the same consolidation key (default: platform + phone + address + city), same shop and an open group join that group; otherwise a new group is created.
- A new group is reserved stock, assigned the least-busy Warehouse Staff and moved to picking.
- If available stock does not cover the group, it is flagged stock shortage and the Store Owner is notified.
- If recipient names in a group differ, Store Owner, Coordinator and the picker are notified.
- A cancelled order re-checks its group; a group without fulfillable orders is cancelled automatically.

3.3.3 Order sync settings

Function trigger: Order Sync Settings screen.

Function description: Admin edits, Store Owner views.

Screen layout:



Function details:

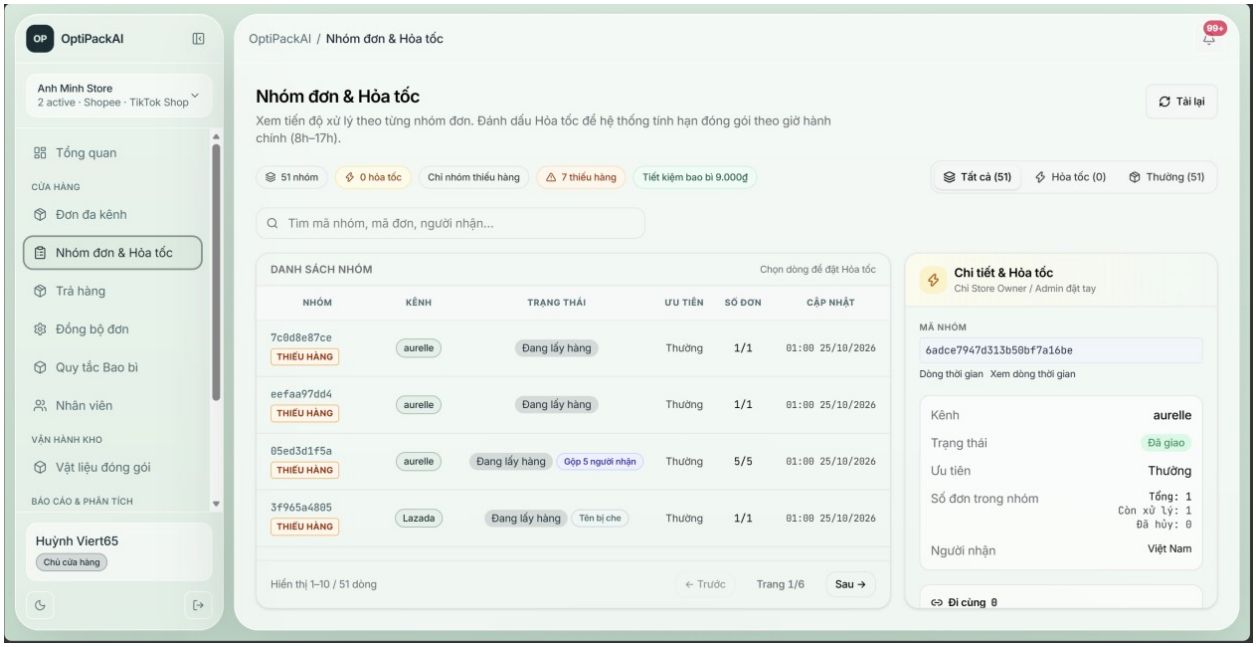
- Every save creates a new version with author and time; history can be viewed.

3.3.4 Manage order groups

Function trigger: Groups & Express screen.

Function description: Store Owner, Shipping Coordinator, Admin.

Screen layout:



Function details:

- Express priority sets a packaging deadline in business hours; warning when less than 1 hour remains.
- Split / merge only while the group is awaiting_packaging or picking with no item scanned; merge requires same shop and same real phone number; a reason is mandatory.

- Seller cancel is allowed only for AURELLE orders not yet handed to the carrier; the customer is refunded if paid.

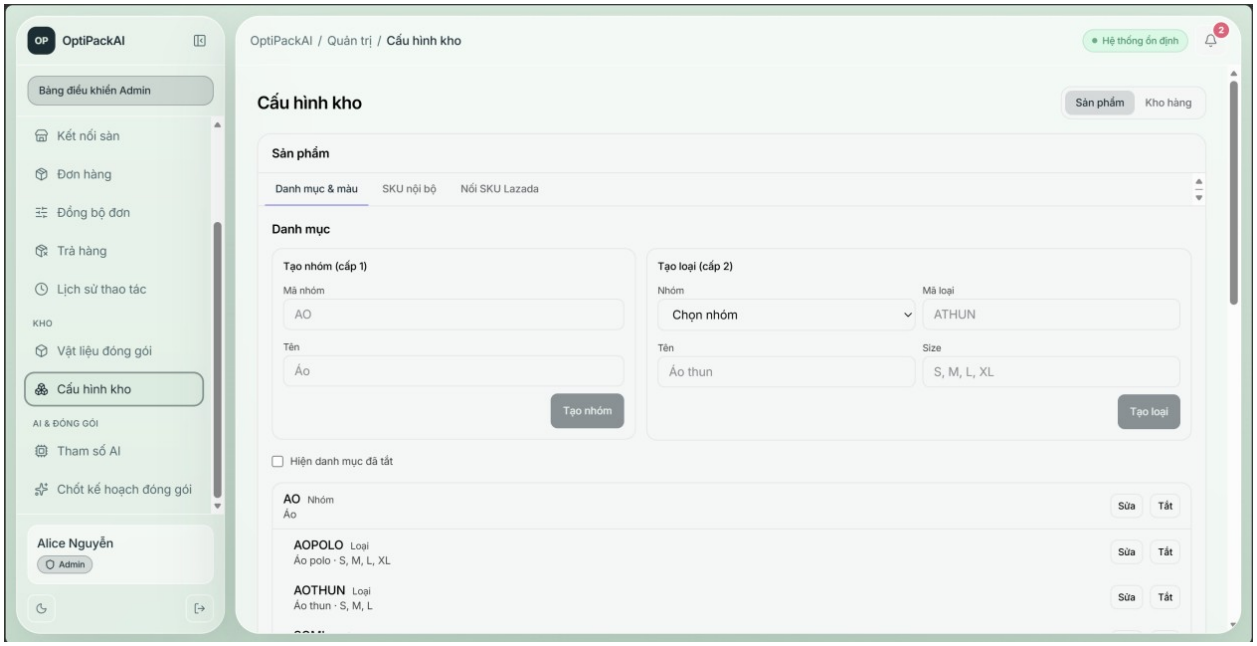
3.4 Warehouse and Inventory

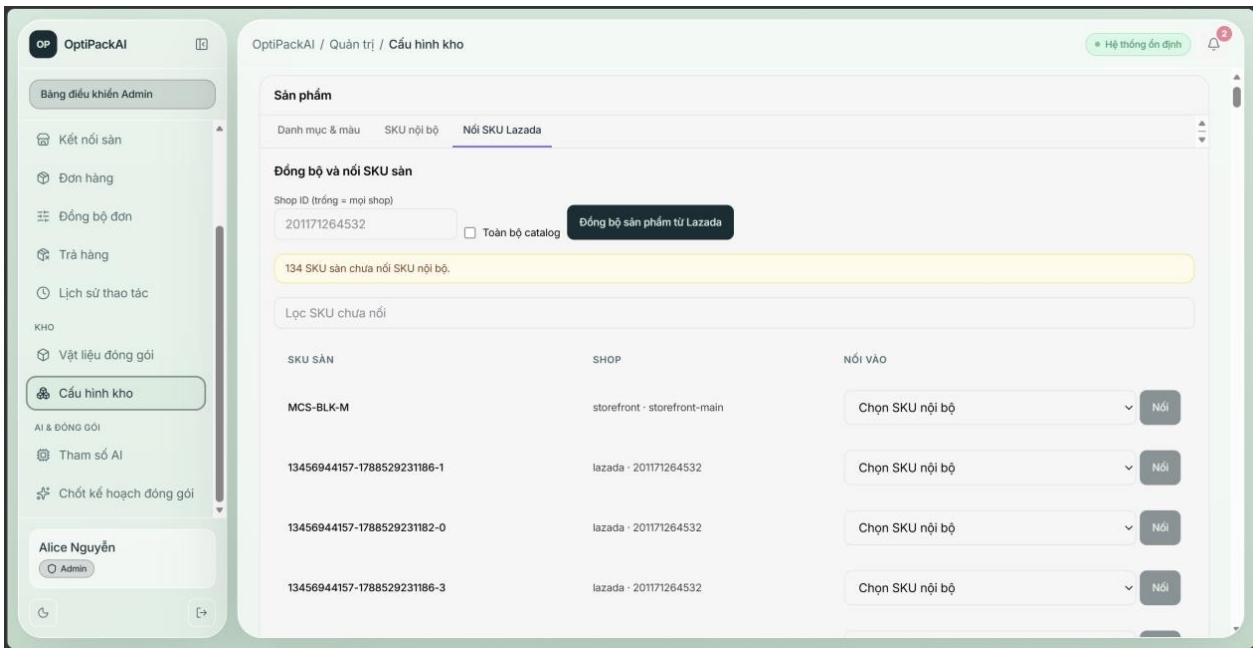
3.4.1 Warehouse structure

Function trigger: Admin Warehouse screen.

Function description: Admin creates warehouses, zones, racks and bins.

Screen layout:





Function details:

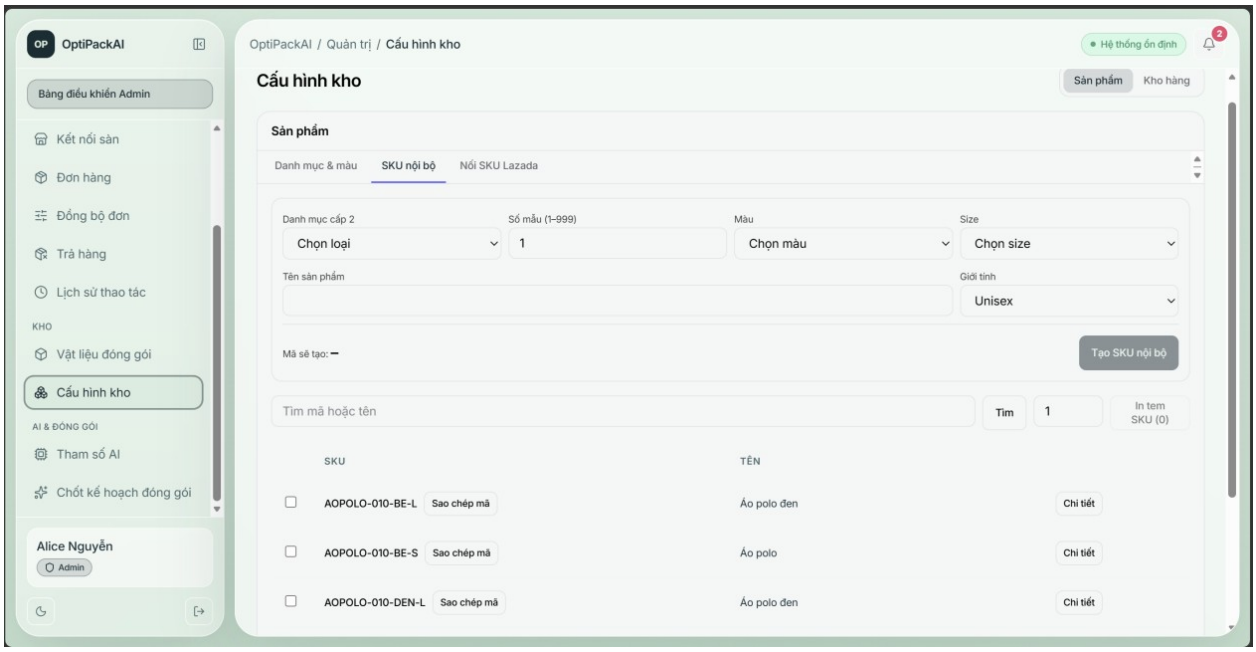
- Bin codes are generated by rack; a bin can have a capacity in units.
- A warehouse, zone or bin can be deleted permanently only when it holds no stock.

3.4.2 Internal SKU and mapping

Function trigger: Admin Warehouse → SKU tab.

Function description: Admin manages internal SKUs and seller-SKU mappings.

Screen layout:



Function details:

- Internal SKU code follows CATEGORY-MODEL-COLOUR-SIZE; category and colour must exist.
- Mapping a seller SKU to an internal SKU pools the stock and moves existing reservations.
- Seller SKUs with the same code as an internal SKU are mapped automatically.

3.4.3 Packaging profile

Function trigger: Product master list → profile form.

Function description: Warehouse Staff confirms dimensions, weight, fragility, item category and folding.

Screen layout:

[Insert screenshot of the "Packaging profile" screen]

Function details:

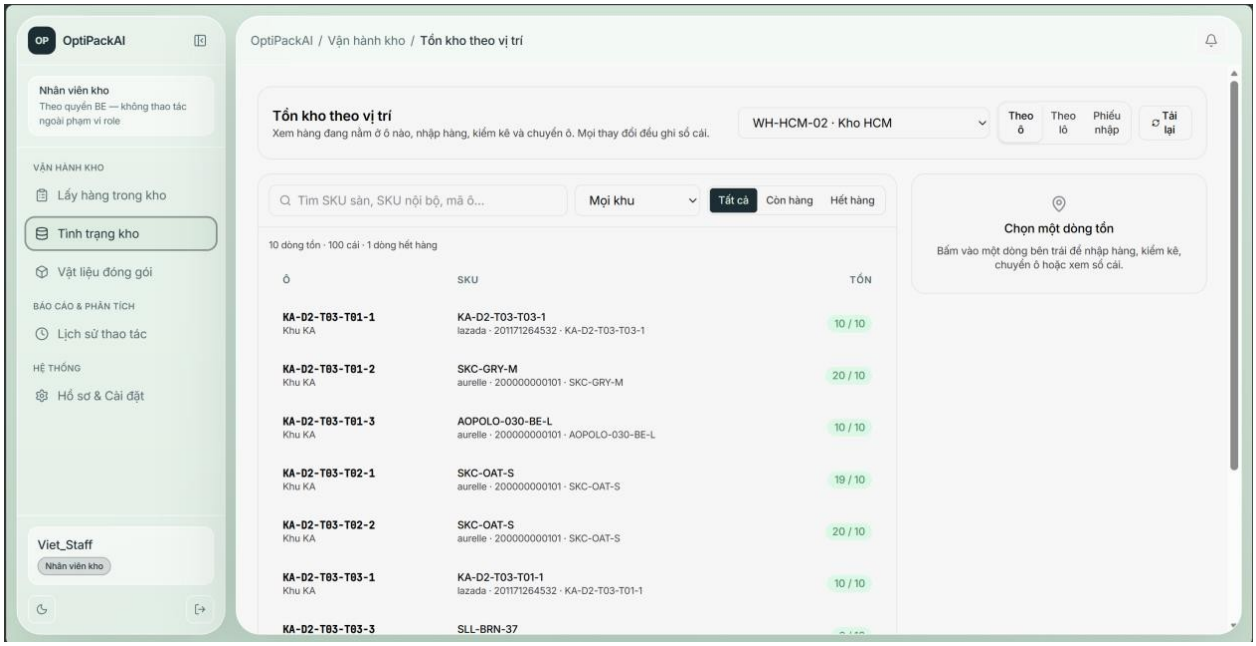
- Only the six clothing categories can be folded in half.
- A group whose SKU has no confirmed profile cannot get a packing plan (packing-blocked).

3.4.4 Bin stock and stock receipt

Function trigger: Warehouse Inventory screen.

Function description: Warehouse Staff assigns SKUs to bins and receives stock by lot.

Screen layout:



Function details:

- Putting a SKU into a second bin of the same warehouse requires explicit confirmation.
- A receipt that exceeds bin capacity is rejected unless forced.
- Each receipt line creates a lot; picking consumes the oldest lot first (FIFO).
- Every change writes an inventory movement; stock-in re-checks groups with shortage.

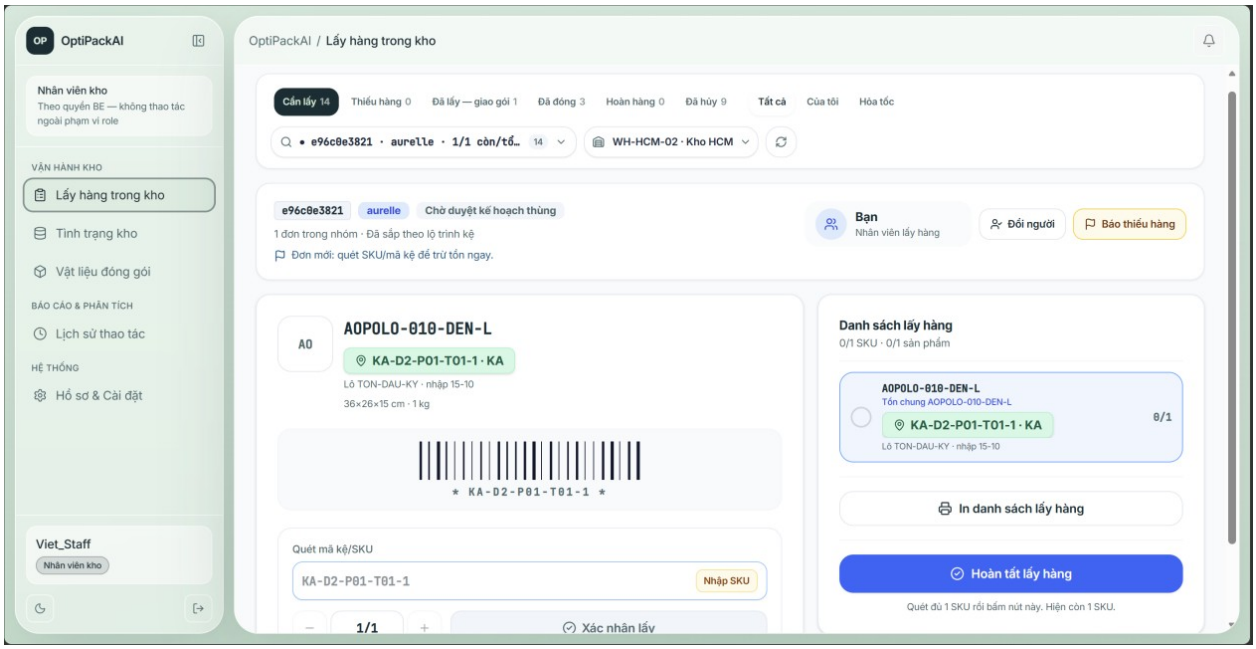
3.5 Picking

3.5.1 Pick items

Function trigger: Warehouse Picking screen → group.

Function description: Warehouse Staff assigned to the group.

Screen layout:



Function details:

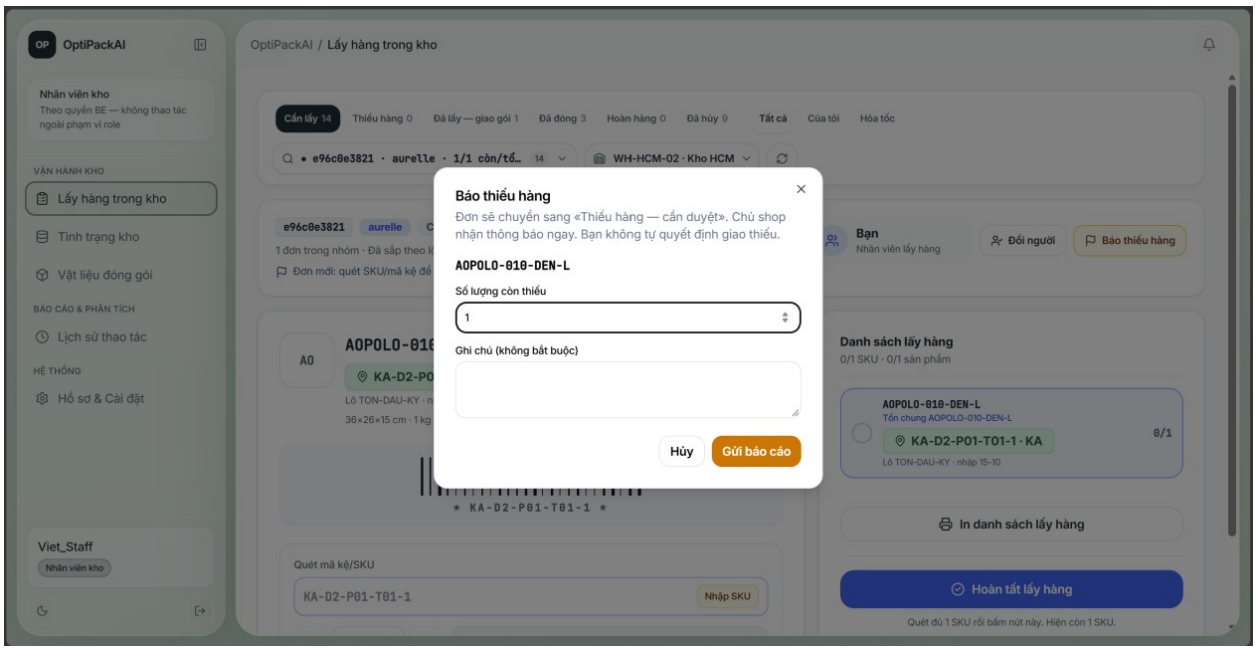
- The picking list shows per SKU: quantity, bin, oldest lot, other bins and split per order for consolidated groups.
- A scan is rejected when the SKU is not in the group, the quantity exceeds the ordered quantity, the warehouse differs or bin stock is insufficient.
- A repeated scan with the same client event id is not counted twice.
- The first scan notifies AURELLE customers "warehouse is preparing your order".
- "Picking completed" is allowed only when every SKU is fully picked.

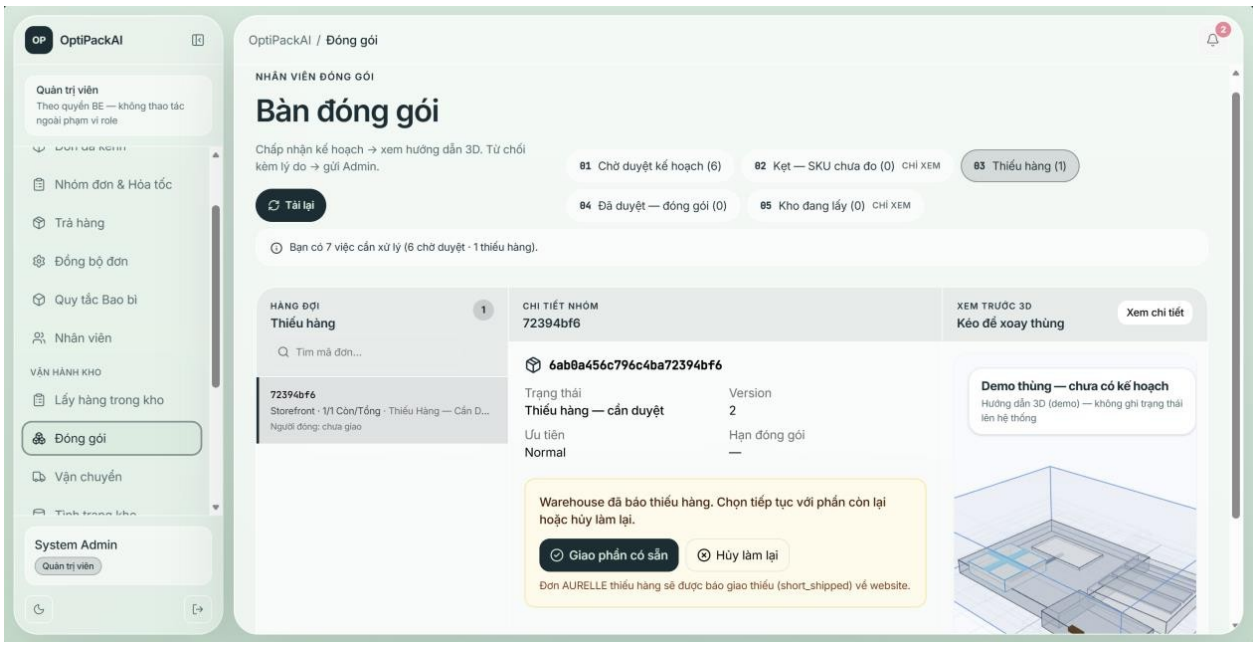
3.5.2 Report missing and decide partial

Function trigger: Picking screen → Report missing; Packing Station → shortage.

Function description: Warehouse Staff reports; Packaging Staff / Admin decides.

Screen layout:





Function details:

- Reporting moves the group to partial_needs_review and notifies Store Owner and Packaging Staff.
- Ship available: group becomes picked, shortage is allocated to oldest orders first and AURELLE customers are informed (not charged).
- Re-pick: a new pick round starts, picked items are returned to their bins and stock is re-reserved.

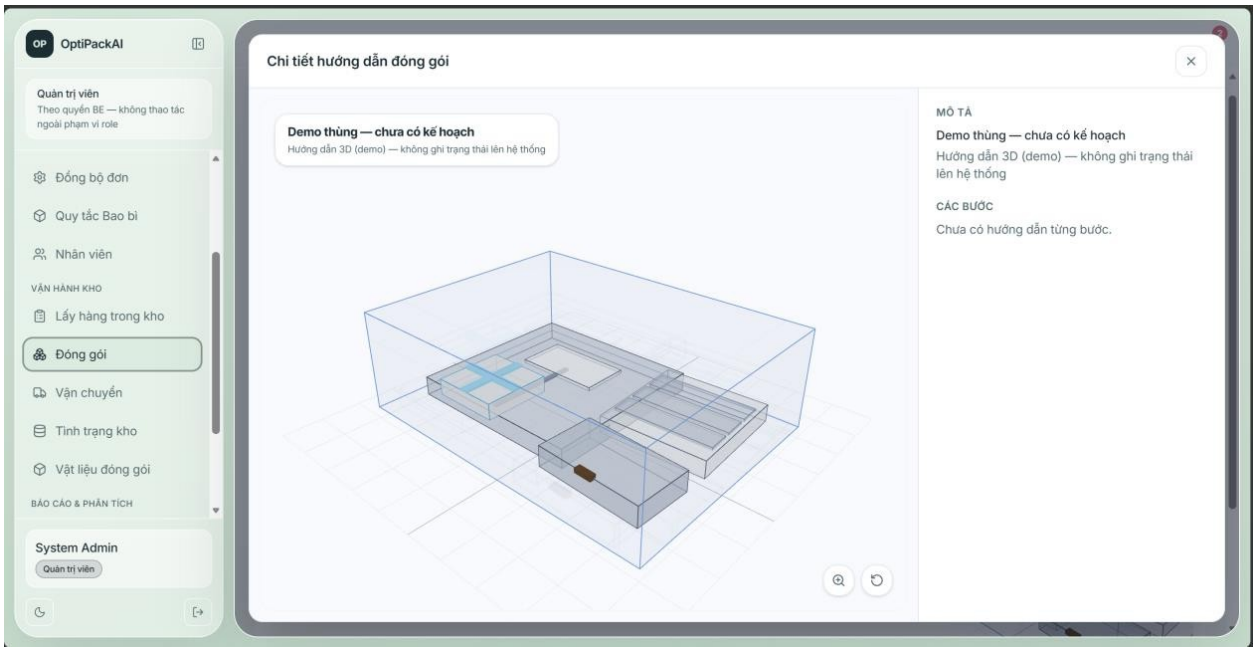
3.6 Packing

3.6.1 Compute and approve packing plan

Function trigger: Automatic; Packing Station.

Function description: System computes; Packaging Staff / Admin approves.

Screen layout:



Function details:

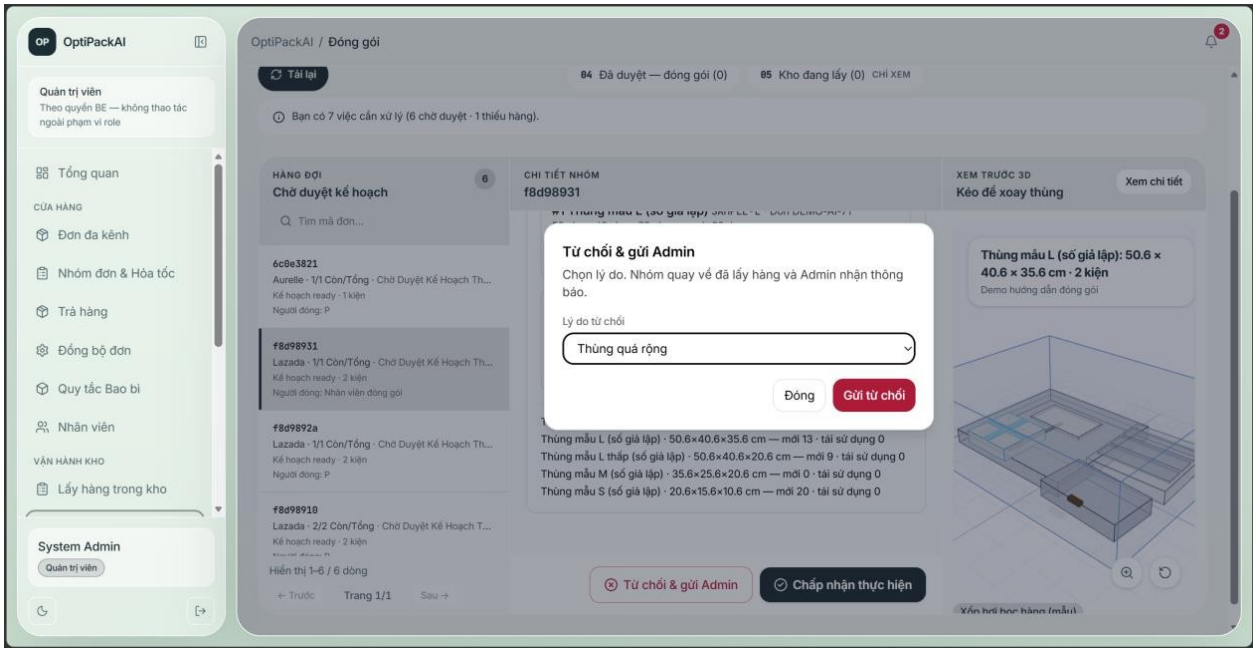
- One plan per order group; each parcel belongs to exactly one order.
- The plan is assigned automatically to the least-busy packer not on leave.
- Approval is possible only when every item of every order is placed.

3.6.2 Reject and recover

Function trigger: Packing Station → Reject.

Function description: Packaging Staff / Admin.

Screen layout:



Function details:

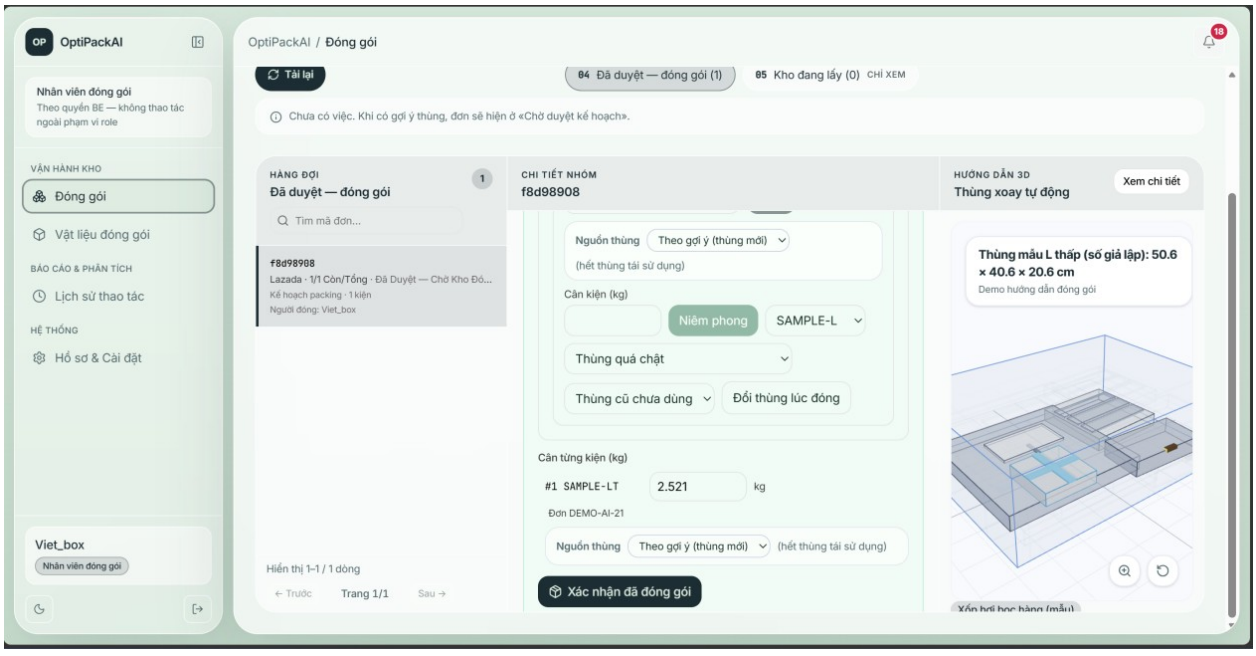
- Reason is chosen from a code list; "Other" needs a note; Admin and Store Owner are notified with a 2-working-hour deadline.
- Recovery: recompute (exclude boxes / prefer cheaper), manual packing approved by another person, or back to picking.

3.6.3 Pack parcels

Function trigger: Packing Station → session.

Function description: Packaging Staff on the Packing Station (Admin too). The back end also accepts the packing-session actions of Warehouse Staff, but the Packing Station screen is not in the Warehouse Staff menu in this release.

Screen layout:



Function details:

- Items must be scanned into the correct parcel; wrong parcel or extra item is blocked.
- Sealing deducts the box (reused shelf first, new box for fragile items) and materials.
- Weight outside tolerance (default 20%) puts the parcel on hold for review by another person.
- "Quick pack" without scanning is blocked when scanning is mandatory or the group has several recipient names.
- The last sealed parcel moves the group to packed and notifies AURELLE customers.

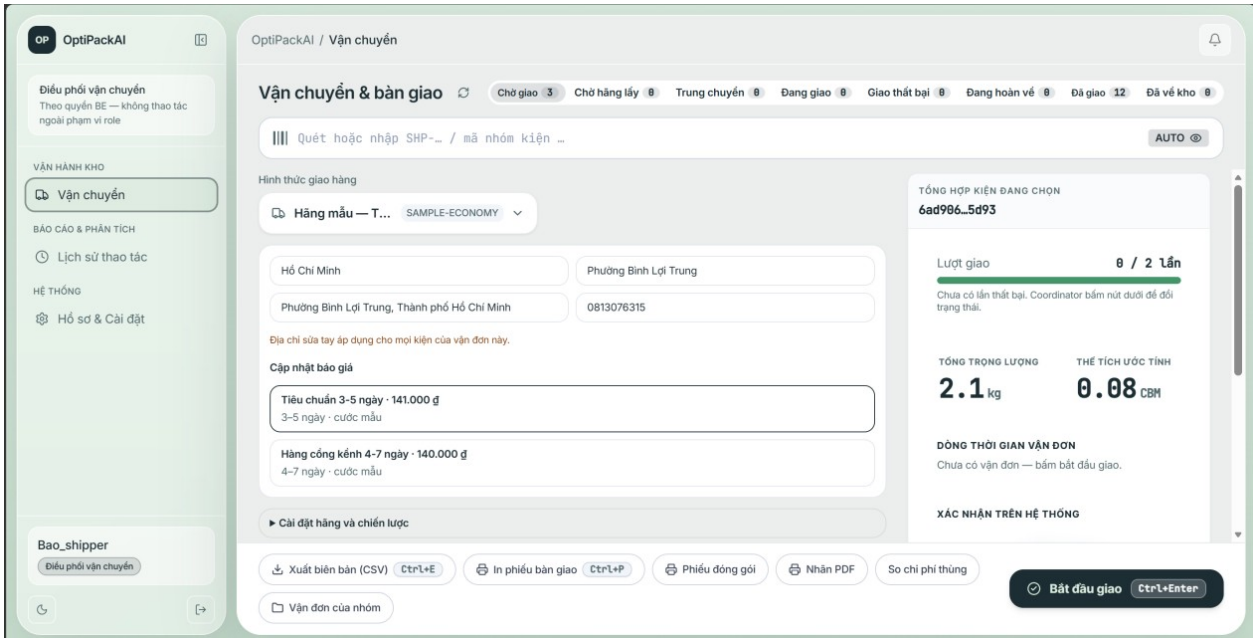
3.7 Shipping

3.7.1 Create shipment

Function trigger: Shipping & Fulfillment screen.

Function description: Shipping Coordinator / Admin.

Screen layout:



Function details:

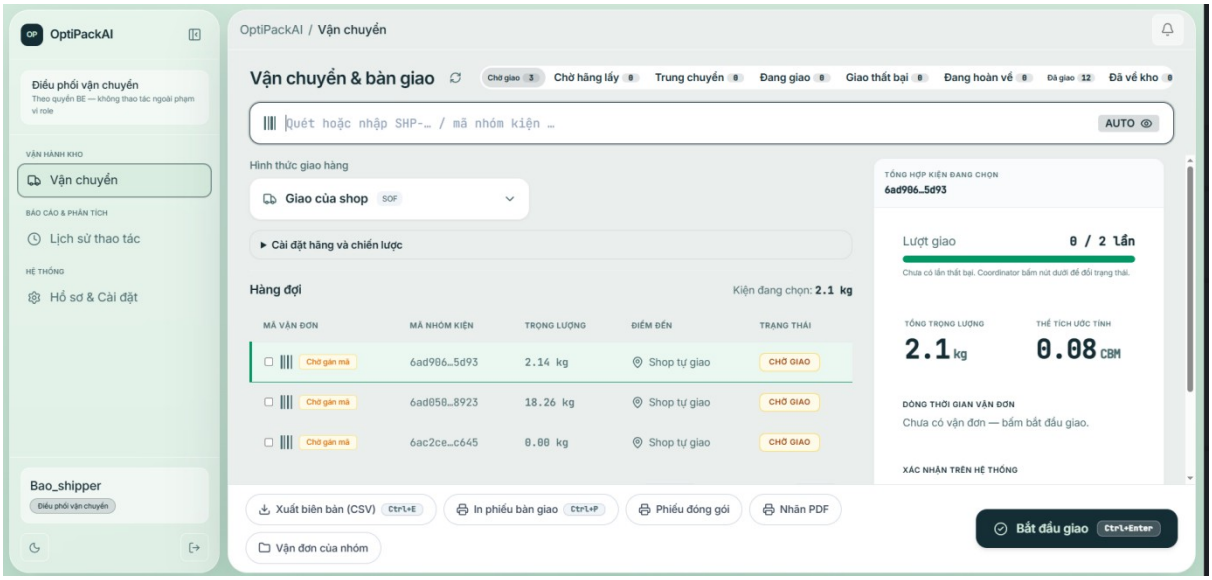
- Only packed groups can be shipped; several groups of the same recipient can be shipped as one trip.
- One GHN order is created per parcel with the recipient of that parcel's order.
- Rejected bookings (e.g. address) are corrected and re-booked without creating duplicates.
- Cost estimation: for each parcel the chargeable weight is max(actual weight, volumetric weight), where volumetric weight = length × width × height of the box / the volumetric divisor of the carrier service (default 5,000); GHN quotes use the fee and converted weight returned by GHN.
- Total fulfilment cost compared for each option = shipping fee of all parcels + box price + filling materials; the default option follows the shipping strategy in Shipping Settings (cheapest, fastest or fixed carrier).
- Each quote shows the expected delivery time returned by the carrier (GHN lead time). Acceptance: the quote list of a packed group shows, per carrier service, the fee, the chargeable weight and the expected delivery time, and with strategy "cheapest" the option with the lowest total cost is selected first.

3.7.2 Track delivery

Function trigger: Automatic (GHN webhook / sync); Shipping screen.

Function description: GHN, Shipping Coordinator.

Screen layout:



Function details:

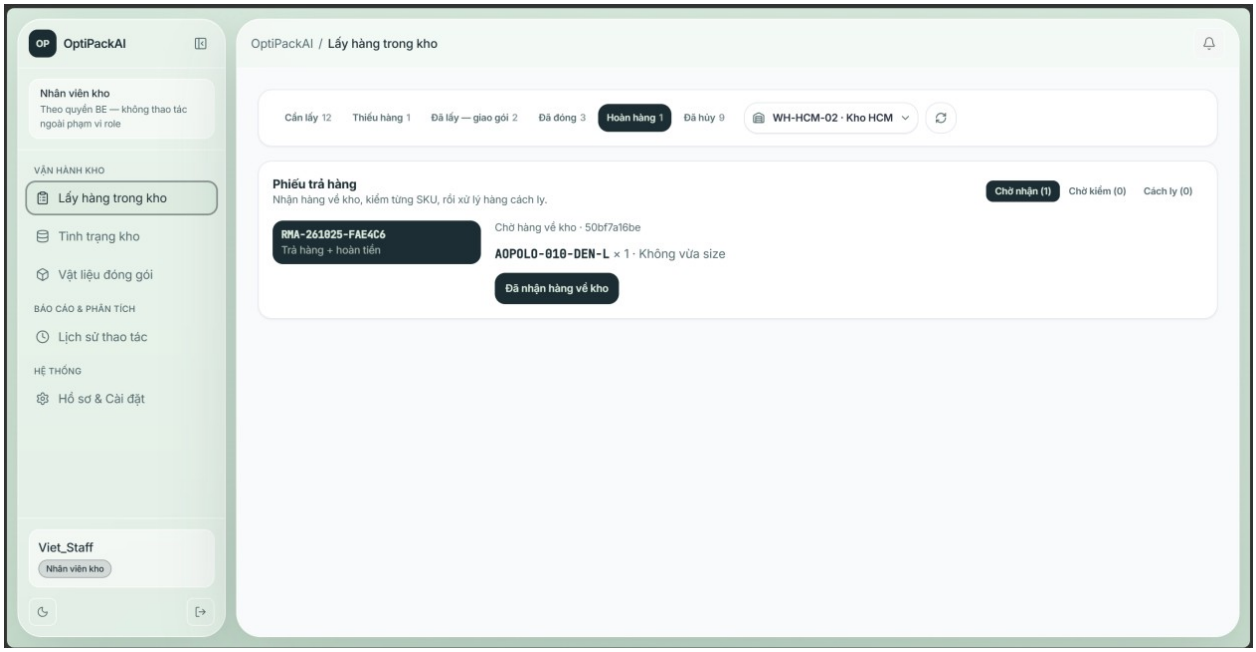
- GHN statuses are mapped to awaiting_pickup, in_transit, out_for_delivery, delivered, delivery_failed, returning_to_warehouse.
- Without carrier API, a second failed attempt or customer refusal automatically returns the shipment.
- Every change is sent to AURELLE so customers see the progress.

3.7.3 Receive returned parcel

Function trigger: Shipping / Warehouse screen → Receive return.

Function description: Warehouse Staff.

Screen layout:



Function details:

- Shipment becomes returned_to_warehouse, group becomes returned and a failed-delivery return ticket is created for inspection.

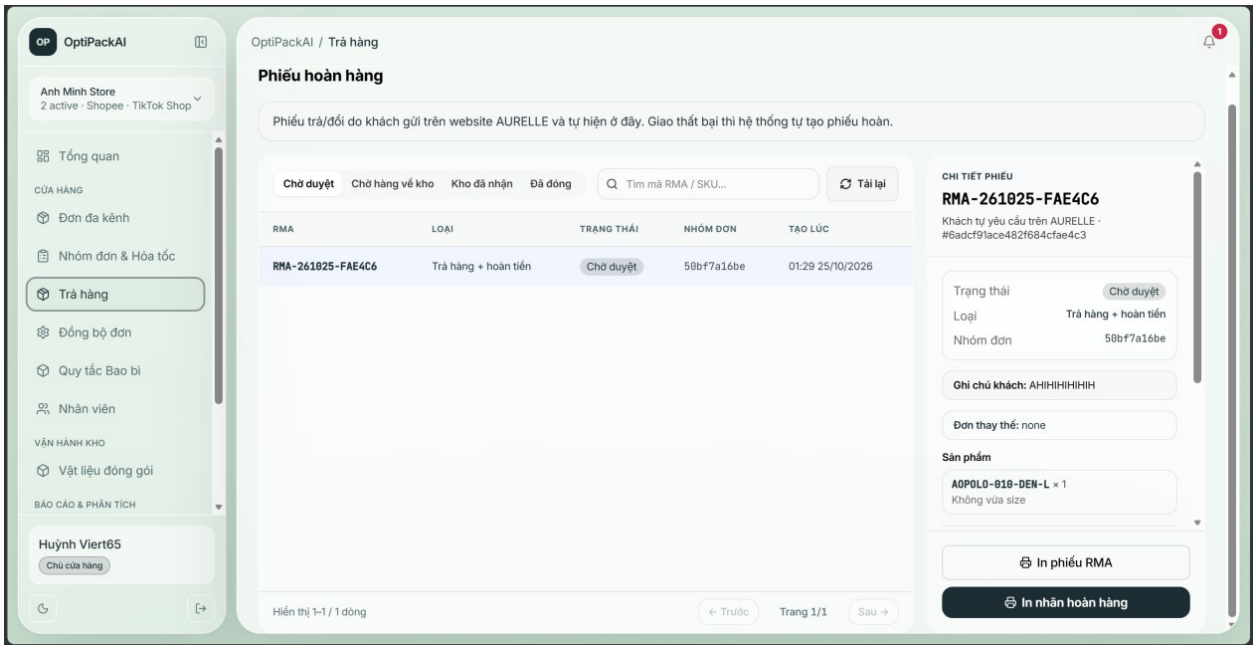
3.8 Returns and Exchanges

3.8.1 Review return ticket

Function trigger: Returns screen.

Function description: Store Owner / Admin.

Screen layout:



Function details:

- The creator of a ticket cannot approve it.
- Reject and cancel require a reason.

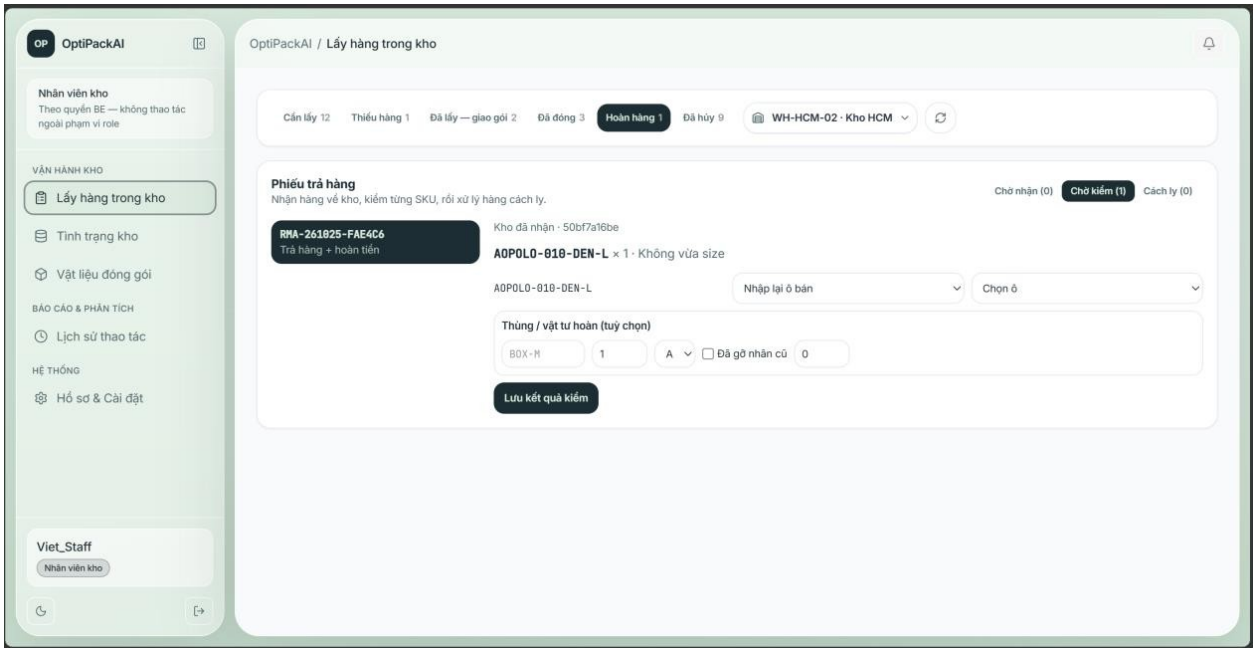
- Refund-only is closed at approval without goods returned.

3.8.2 Receive and inspect

Function trigger: Warehouse Picking screen → Returns tab (Warehouse Staff); Returns screen (Admin).

Function description: Warehouse Staff / Admin.

Screen layout:



Function details:

- Each SKU is restocked, quarantined or discarded; restocked items return to a bin and write the ledger.
- Exchange tickets create a replacement order EXC-… (price 0) that goes through picking, packing and shipping.

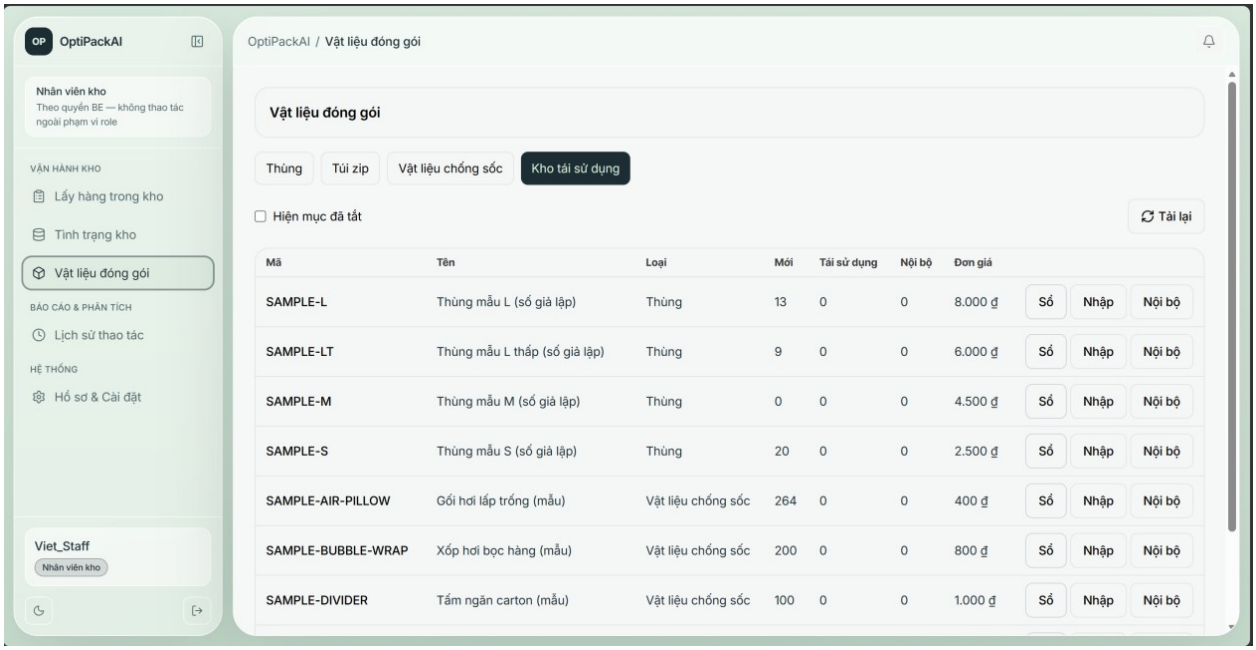
3.9 Packaging Materials

3.9.1 Manage packaging catalog and movements

Function trigger: Packaging Materials / Rules screens.

Function description: Admin manages; Warehouse Staff records stock-in and use; Store Owner views savings.

Screen layout:



Function details:

- Boxes have new and reused stock; fragile items use new boxes unless allowed in settings.
- Every purchase, internal use, consumption and recovery is recorded as a movement.

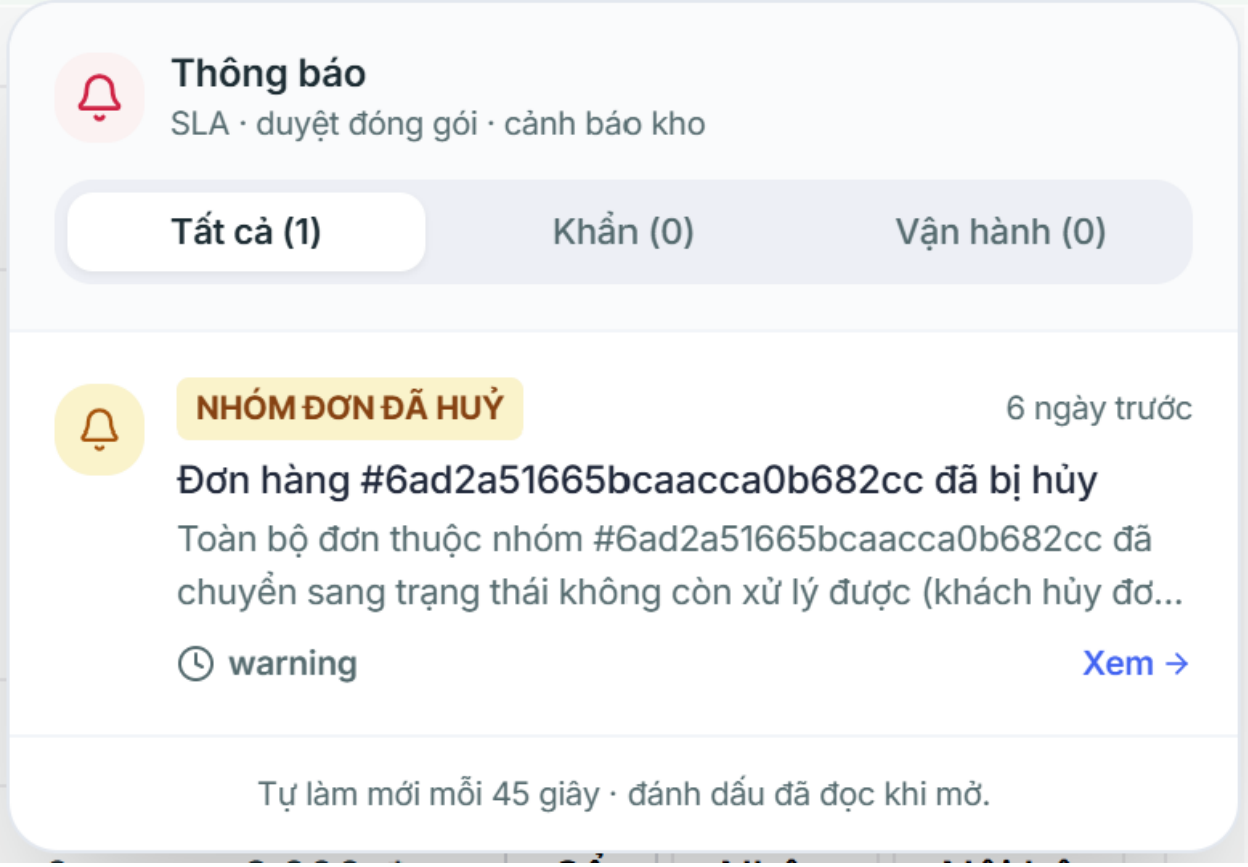
3.10 Notifications

3.10.1 Notifications

Function trigger: Top bar bell.

Function description: All staff roles.

Screen layout:



Function details:

- Notifications are addressed to a role or to a user and link to the related entity.

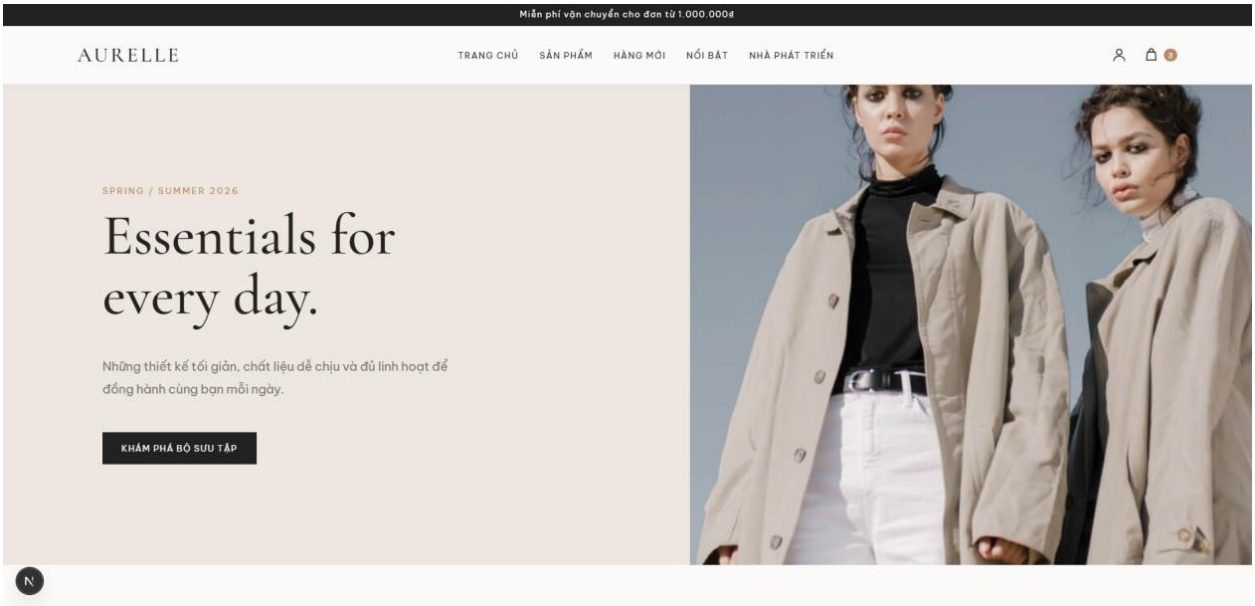
3.11 AURELLE Customer Website

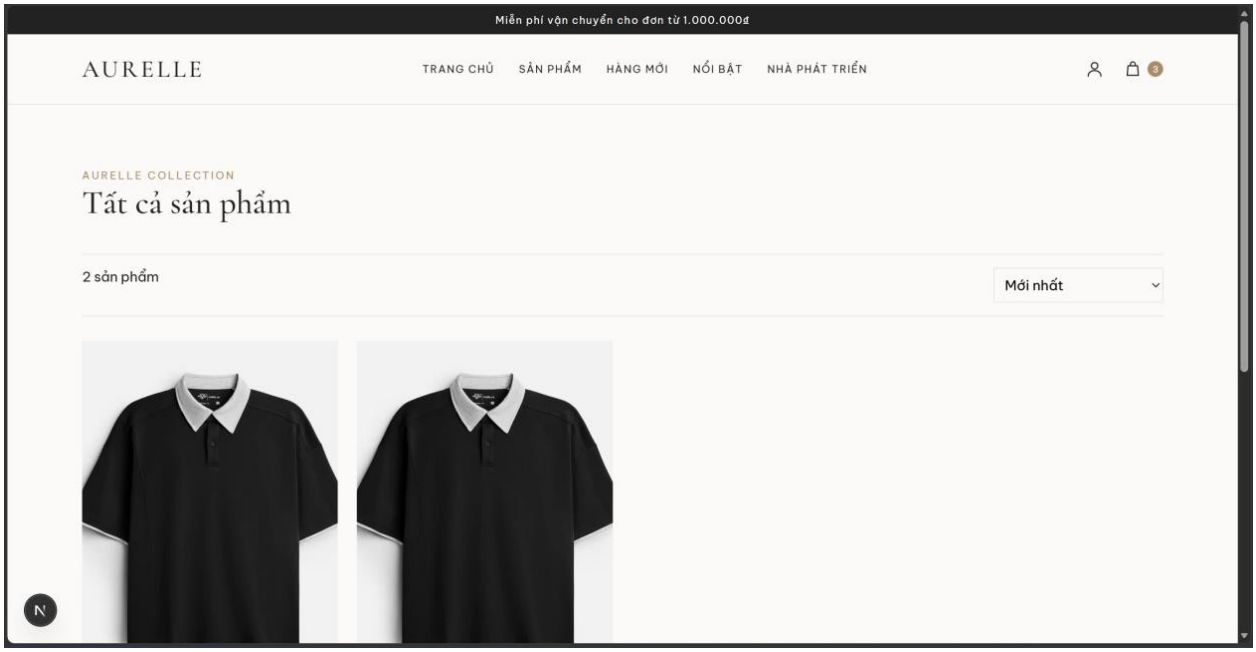
3.11.1 Browse, cart and checkout

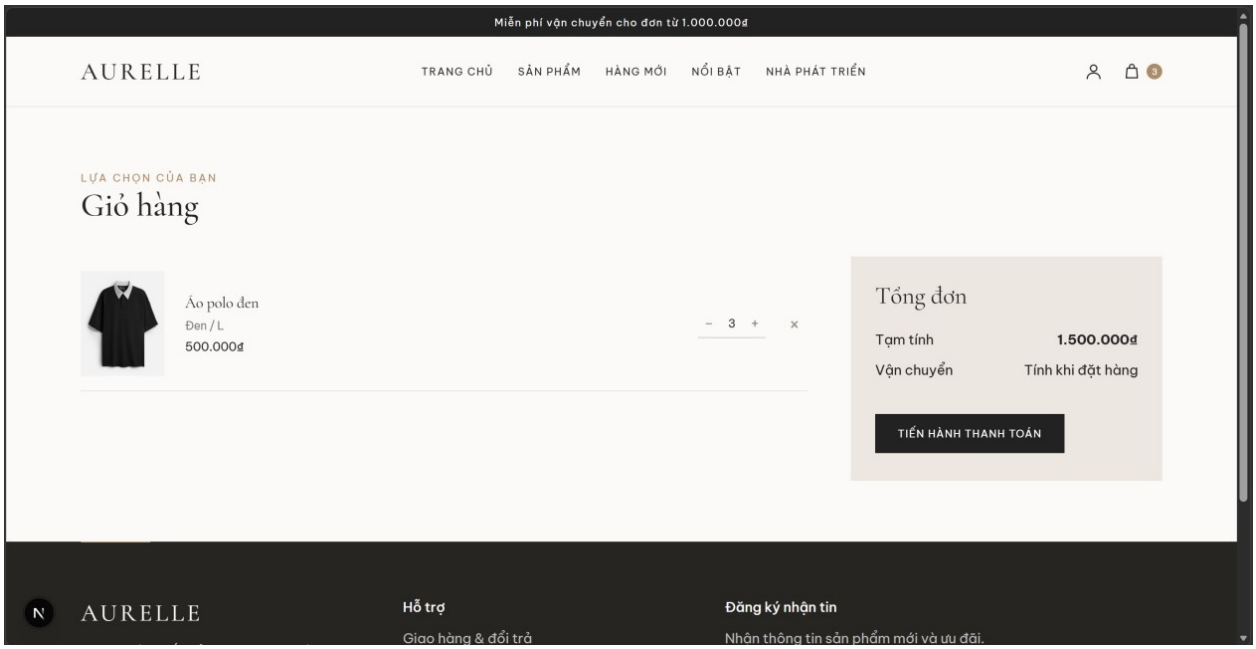
Function trigger: Home → Shop → Product → Cart → Checkout.

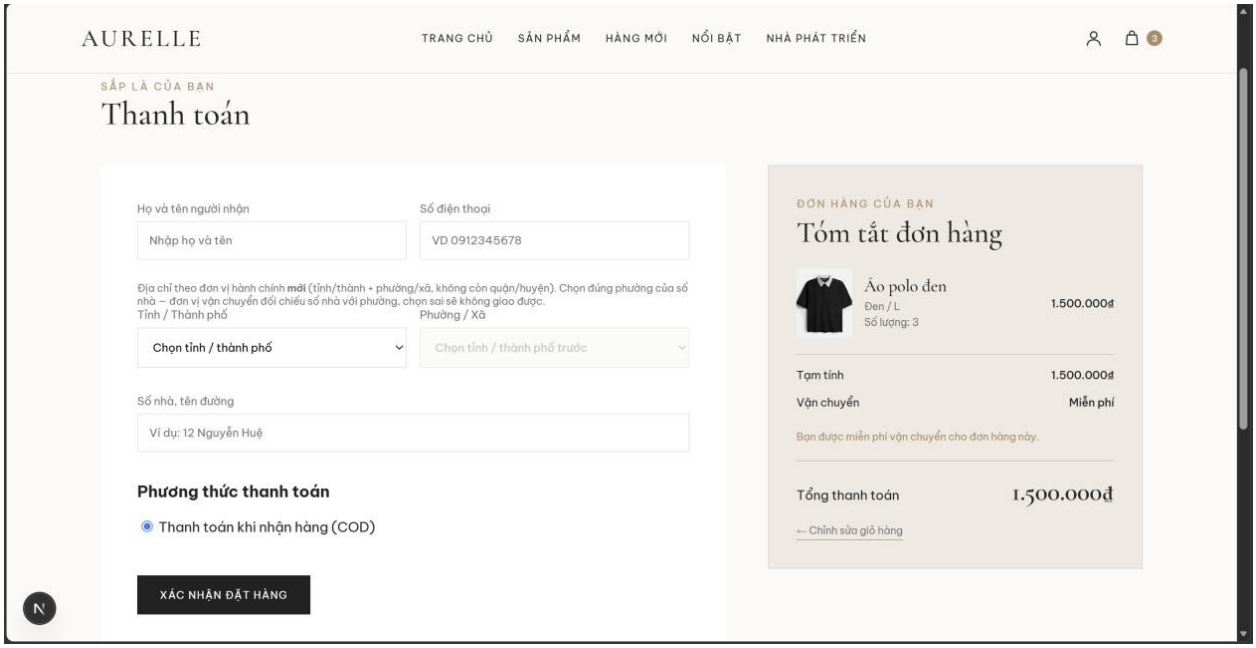
Function description: Customer.

Screen layout:




Miễn phí vận chuyển cho đơn từ 1.000.000đ
AURELLE
TRANG CHỦ
SẢN PHẨM
HÀNG MỚI
NỔI BẬT
NHÀ PHÁT TRIỂN
AURELLE COLLECTION
Tất cả sản phẩm
2 sản phẩm
Mới nhất


Miễn phí vận chuyển cho đơn từ 1.000.000đ
AURELLE
TRANG CHỦ
SẢN PHẨM
HÀNG MỚI
NỔI BẬT
NHÀ PHÁT TRIỂN
LỰA CHỌN CỦA BẠN
Giỏ hàng
Áo polo đen
Đen / L
500.000đ
Tổng đơn
Tạm tính
1.500.000đ
Vận chuyển
Tính khi đặt hàng
TIẾN HÀNH THANH TOÁN
AURELLE
Hỗ trợ
Giao hàng & đổi trả
Đăng ký nhận tin
Nhận thông tin sản phẩm mới và ưu đãi.



Function details:

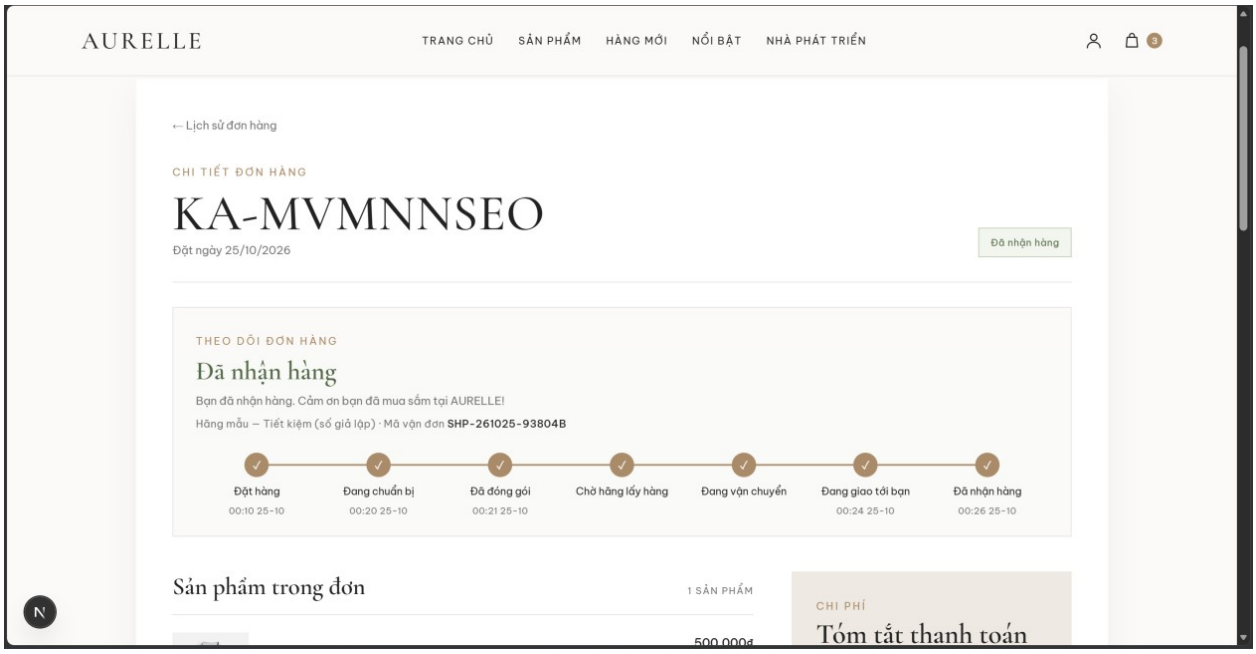
- Only variants whose stock is managed by the OptiPack warehouse are sellable.
- Address uses the new two-level administrative units (province, ward); phone must be 10 digits starting with 0.
- Payment method: COD or bank transfer.

3.11.2 Track and cancel order

Function trigger: My Orders → Order Detail.

Function description: Customer.

Screen layout:



Function details:

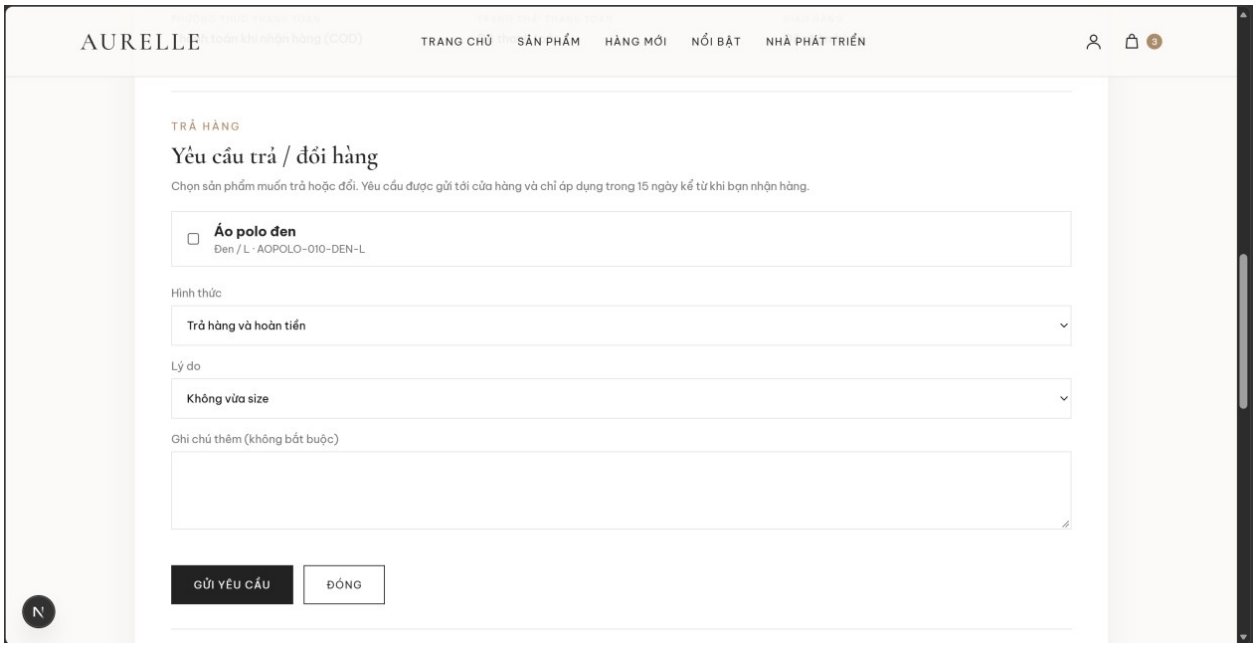
- A 7-step progress bar (Placed, Preparing, Packed, Waiting for pickup, In transit, Out for delivery, Delivered) is shown from the moment the order is placed, with branches for cancellation, failed delivery and return.
- The page refreshes every 30 seconds while the order is not finished.
- The customer can cancel only while the order is pending and not packed.

3.11.3 Return and exchange request

Function trigger: Order Detail → Request return.

Function description: Customer.

Screen layout:



Function details:

- Allowed within 15 days after delivery, one open request per order; exchange target must be in stock.
- A pending request can be cancelled by the customer.

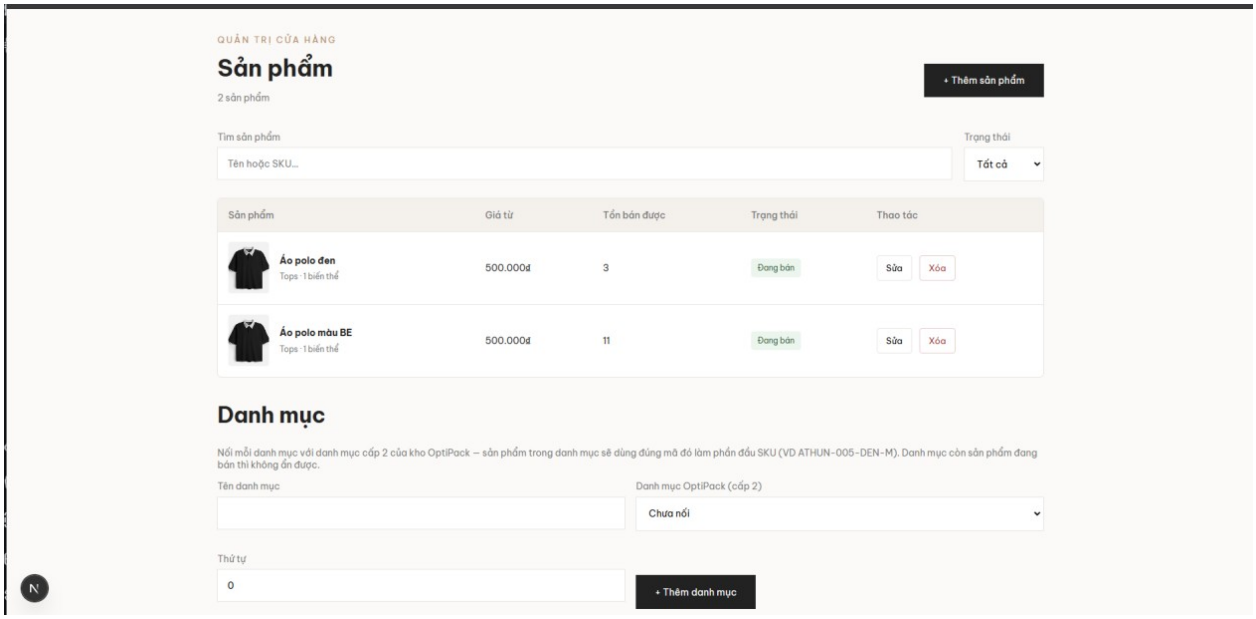
3.12 AURELLE Web Administration

3.12.1 Manage storefront catalog

Function trigger: AURELLE Web Admin page.

Function description: AURELLE Web Admin.

Screen layout:



Function details:

- New variants must use OptiPack category, model number, colour and size so the SKU matches an internal SKU.
- Saving a product notifies OptiPack (product-updated webhook) so the catalog is synchronized.

4. Non-Functional Requirements

4.1 External Interfaces

Interface	Description
User interfaces	Web portal for staff (web browser; a mobile scanning app is planned for Iteration 3 and is not part of this release), AURELLE customer website and admin page (desktop and mobile browser).
Lazada Open Platform	HTTPS REST with OAuth 2.0 access / refresh tokens and request signing.
AURELLE Open API	HTTPS REST in Lazada style with HMAC-SHA256 signed requests; webhooks signed with the app secret. In the demo the Open API runs as a simulator (be/scripts/aurelle-mock-server.ts, port 4000) that reads the AURELLE website data.
GHN API	HTTPS REST with token and shop id; webhooks for status changes; A5 shipping labels.
Google OAuth 2.0	Login for existing staff accounts.
SMTP (Gmail)	Transactional e-mails.
CP-SAT packer service	Internal HTTP service (Python, OR-Tools) used to prove packing plans; optional.
AI text service	Groq API (OpenAI-compatible chat completions, model openai/gpt-oss-120b, timeout 20 s) used to word packing guides; the template text is used when it is unavailable. No customer data is sent.

4.2 Quality Attributes

4.2.1 Usability

- Each role lands on its own home screen and sees only its menu items.
- Every error returned to the user has a stable error code and a readable Vietnamese message.
- Picking and packing screens on the web portal accept barcode scanner input and manual entry.
- Metric: a new staff member finishes the main task of the own role (pick one order group, pack one parcel, create one shipment) without help in ≤ 5 minutes after a 15-minute

introduction; at least 4 of 5 test users succeed. Measurement: observed usability session with a stopwatch during the Verification stage (Report 5).

- Metric: every error message shown to the user is in Vietnamese and comes with an error_code. Measurement: checked on every screen of section 3.1.2 during system testing (Report 5).

4.2.2 Reliability

Requirement	Metric / threshold	How it is achieved / measured
Stock never goes negative or double-counted	0 negative bin stock	Atomic conditional decrement in a MongoDB transaction; idempotent scans by client event id; unit tests.
No duplicated external events	0 duplicates	Processed webhook / carrier event collections and idempotency keys on write-back.
Status write-back to AURELLE	Retried up to 8 times with back-off 30 s → 30 min	Outbox written in the same transaction as the business change.
Degraded mode when packer service is down	Plans still produced	Fallback to the built-in BRKGA solver (label "heuristic").
Automated regression	≥ 1,490 automated back-end tests (150 spec files on 08/11/2026) pass on every push	GitHub Actions CI: type check, lint, tests, builds.

4.2.3 Performance

Requirement	Metric / threshold	How it is achieved / measured
New Lazada orders appear in OptiPack	≤ configured sync interval + 1 minute (default ≤ 6 minutes)	Scheduler every minute; each shop is synchronized when its interval (default 5 minutes) has passed.
Picked group gets a packing plan	Job runs every 10 s; BRKGA p95 ≤ 1 s per order; CP-SAT limit 6 s per order	Benchmark be/scripts/solver-benchmark.ts on 300 orders (last run p95 560 ms).
Customer sees status change on AURELLE	≤ 15 s write-back + ≤ 30 s page refresh	Write-back job every 15 s; page auto-refresh.
AURELLE stock follows warehouse stock	≤ 1 minute	Stock push every minute.
API protection	600 requests / minute / client	Global rate limiter.

4.2.4 Security

- Passwords hashed with bcrypt; account lockout after 5 failed logins for 15 minutes.
- JWT access token 24 hours, refresh token 30 days with rotation and reuse detection; logout revokes the refresh token.
- Optional two-factor authentication (TOTP + 10 one-time backup codes) for every staff account; trusted device 30 days.
- Marketplace tokens encrypted with AES-256-GCM; AURELLE and GHN callbacks verified by signature / secret.
- Role-based access control on every endpoint; HTTP security headers (Helmet); login audit log.
- Measurement: back-end unit tests of the auth module (auth.service, mfa.service and jwt.strategy specs) run in CI on every push; during system testing each role opens the screens and endpoints outside its permissions in section 3.1.3 and must receive 403 (100% of cases); 5 wrong passwords must lock the account for 15 minutes.

5. Requirement Appendix

5.1 Business Rules

ID	Rule Definition
BR-01	Orders of the same shop with the same consolidation key are grouped while the group is still awaiting_packaging or picking (configurable).
BR-02	Each parcel contains items of exactly one order; each parcel has its own carrier tracking code with the recipient of that order.
BR-03	Split or merge of order groups is allowed only before the first item is scanned; merge requires same shop and same real phone number.
BR-04	Stock is reserved per order group; picking consumes the oldest lot first (FIFO).
BR-05	A group can be marked picked only when all ordered quantities are scanned; otherwise missing items must be reported.
BR-06	A packing plan cannot be approved while any item is unplaced; a manual plan must be approved by another person.
BR-07	A parcel whose weight differs from the estimate by more than the tolerance (default 20%) is held and reviewed by a different person.
BR-08	Fragile items are packed in new boxes unless reused boxes are allowed in packing settings.
BR-09	A rejected packing plan must be handled within 2 working hours (08:00–17:00).
BR-10	Without carrier API, a shipment returns to the warehouse after 2 failed delivery attempts or customer refusal.
BR-11	Return requests are accepted within 15 days after delivery; one open request per order.
BR-12	The creator of a return ticket cannot approve it; reject and cancel require a reason.
BR-13	A customer can cancel an AURELLE order only while it is pending and not packed; the seller can cancel until it is handed to the carrier.
BR-14	Short-shipped items are allocated to the oldest orders first and are not charged.
BR-15	Only the six clothing categories can be folded in half in a packing plan.
BR-16	Staff on leave are not assigned work automatically or manually.
BR-17	A new account must change its temporary password within 72 hours.

5.2 Common Requirements

- Every state-changing request that can be performed by two people at the same time carries the version of the record (expected_version); an outdated version is rejected with a conflict error.
- Errors are returned as { error_code, message, details } with HTTP status 400 / 401 / 403 / 404 / 409 / 422.
- All dates are stored in UTC and displayed in Asia/Ho_Chi_Minh time.
- Every business action that changes the state of an order group, return ticket or stock is recorded in the history.

5.3 Application Messages List

Code	Type	Context	Content
AUTH_INVALID_CREDENTIALS	Error	Login	Email hoặc mật khẩu không chính xác. (Wrong e-mail or password.)
AUTH_ACCOUNT_LOCKED	Error	Login	Account locked for 15 minutes after too many failed attempts.
AUTH_PASSWORD_DEADLINE_LOCKED	Error	Login	Account locked: temporary password not changed within 72 hours.
ORD_GROUP_STATE_CONFLICT	Error	Any group action	The group was changed by someone else — reload and try again.
ORD_GROUP_PICK_EXCEEDS_ORDERED	Error	Picking	Scanned quantity exceeds the ordered quantity.
ORD_GROUP_INSUFFICIENT	Error	Picking	Not enough stock in the bin — report missing

Code	Type	Context	Content
_STOCK			items.
ORD_GROUP_PICK_INCOMPLETE	Error	Picking	Some SKUs are not fully picked.
ORD_GROUP_WRONG_WAREHOUSE	Error	Picking	This round is being picked in another warehouse.
ORD_GROUP_REGROUP_HAS_PICKS	Error	Split / merge	Items already scanned — the group cannot be split or merged.
ORD_GROUP_MERGE_RECIPIENT_MISMATCH	Error	Merge	Different phone numbers — use "Link" instead of merging.
PACKING_SCAN_REQUIRED	Error	Quick pack	Scanning each item is required (setting or several recipients).
PACKING_PACK_WEIGHTS_MISMATCH	Error	Packing	Weights must be entered once for every unsealed parcel.
PKG_BOX_OUT_OF_STOCK	Error	Packing	The selected box is out of stock.
WH_SKU_IN_OTHER_BIN	Confirmation	Bin assignment	The SKU is already in another bin — confirm to use a second bin.
RMA_WINDOW_EXPIRED	Error	Returns	More than 15 days since delivery — return period expired.
RMA_SIMULATION_DISABLED	Error	Returns	Simulated returns are disabled; customers request returns on AURELLE.
PM_FOLD_NOT_ALLOWED	Error	Packaging profile	Only clothing categories can be folded in half.
consolidation_name_mismatch	Notification	Order groups	Consolidated group has several recipients with different names.
stock_shortage	Notification	Order groups	Available stock does not cover the group.
missing_item	Notification	Picking	Warehouse Staff reported a missing item.
packing_blocked	Notification	Packing	Packing plan blocked — SKU needs measurement.
delivery_failed	Notification	Shipping	Carrier reported a failed delivery.
return_requested	Notification	Returns	A customer requested a return / exchange.

5.4 Other Requirements

The main business workflows WF-01 to WF-06 that combine the features above (actors, triggers, outcomes and state changes) are specified with activity diagrams in Report 4 – Software Design Document.